

Visual Perception – Lecture 1

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Why study visual perception?

- Hypothetical project
 - Design a device that can **locate, describe, and identify** all objects in the environment, including their distance from the device and the spatial relationship to each other.
 - Additionally, make the device capable of **traveling** from one point to another (avoiding obstacles), **evaluating and reflecting on** what is processed, having a **conscious experience** of what is processed, **sorting information** into relevant categories, and **storing details** for future encounters.

Build a '**perception machine**' ...
...such as a human.

Perception determines behaviour!

Why study visual perception?

Perception determines behaviour

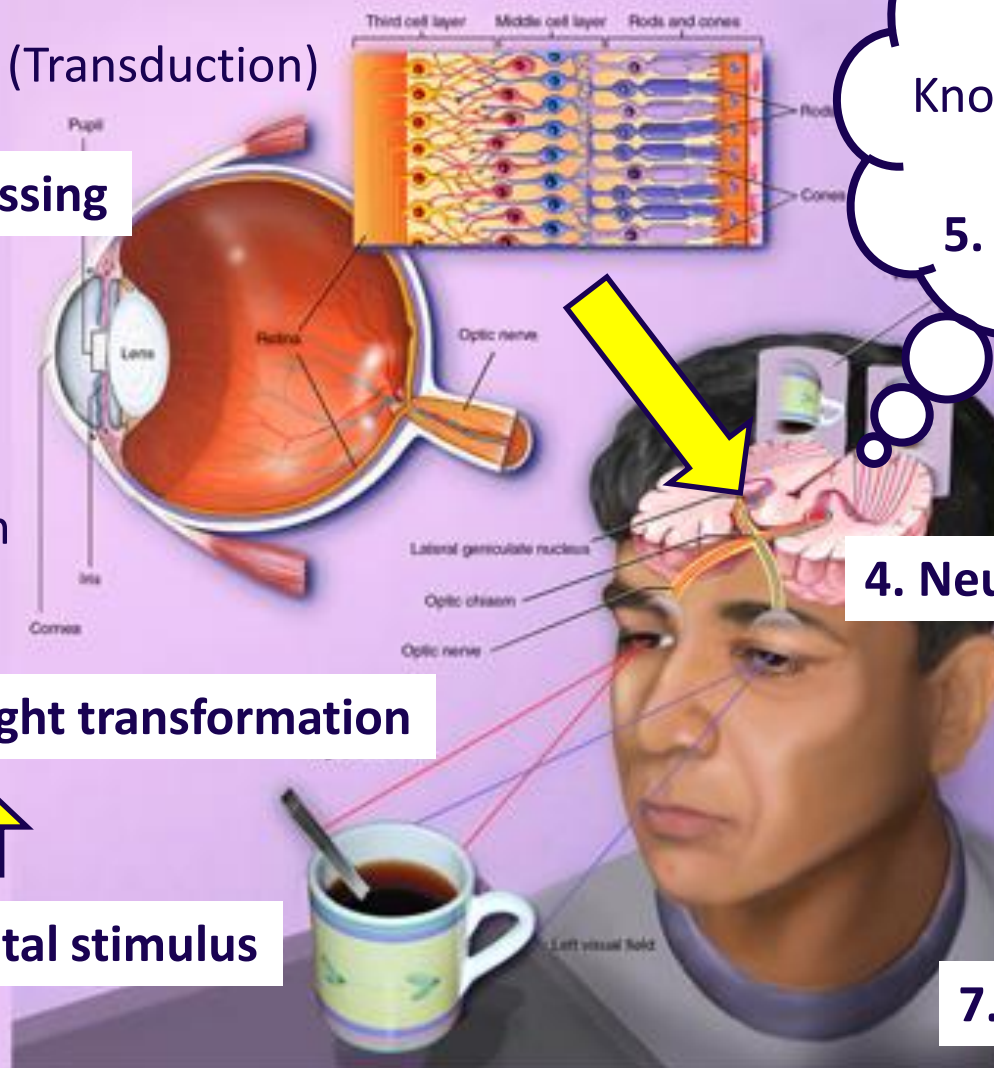
2nd transformation (Transduction)

3. Receptor processing

1st transformation
& Representation

2. Light transformation

1. Environmental stimulus



4. Neural processing

5. Perception

6. Recognition

7. Action

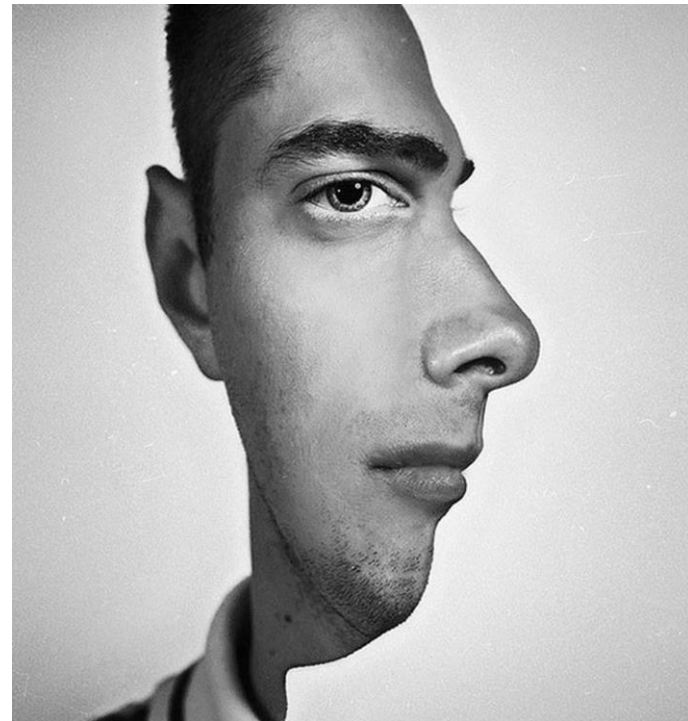
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- You would like to study Psychology and need to **pass the Intro 1 exam** to get into Year 2.



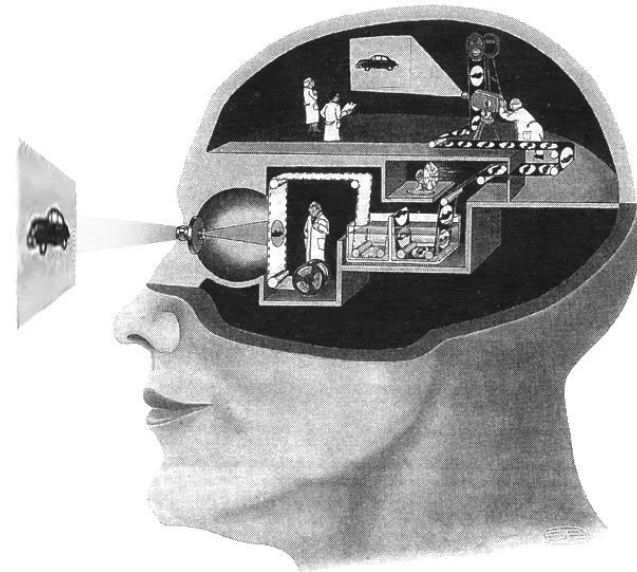
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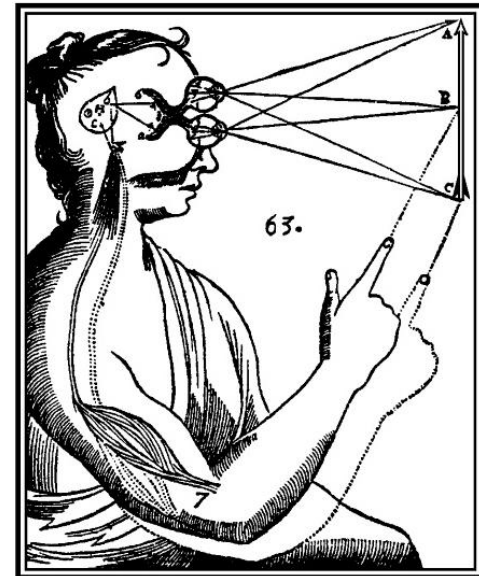
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- You would like to learn more about one of the **oldest scientific disciplines** in Psychology.



Why study visual perception?

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- You would like to be able to **answer questions from other experts**, e.g., Why do I super-duper engineer struggle so badly with building this fancy perception machine?
- You would like to learn more about one of the **oldest scientific disciplines** in Psychology.
- You are a true psychologist and want to **learn everything about the (human) mind and behaviour**.



Intro to Visual Perception - Overview

1. Basics of vision I: The eye
2. Basics of vision II: The brain



Week 1
Physiology

Intro to Visual Perception - Overview

1. Basics of vision I: The eye
2. Basics of vision II: The brain
3. The psychology of visual perception
4. When perception goes wrong

Week 1
Physiology

Week 2
Models, theories &
tests

Intro to Visual Perception - Overview

1. Basics of vision I: The eye
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5. Colour perception
6. Depth/Size perception
7. Motion perception
8. Object perception and recognition

Week 1
Physiology

Week 2
Models, theories &
tests

Weeks 3&4
Features & objects

Intro to Visual Perception - Overview

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8. Object perception and recognition
9. Interim revision
10. Formative class test and discussion

Week 1
Physiology

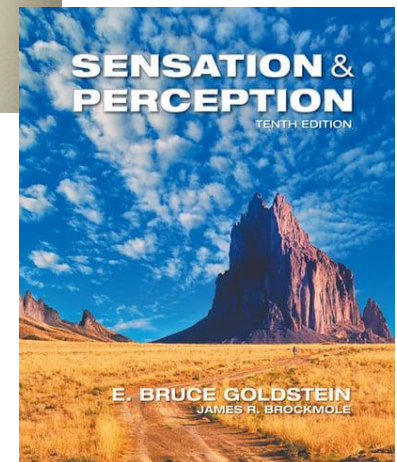
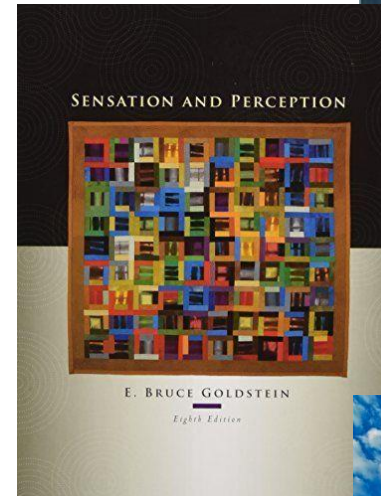
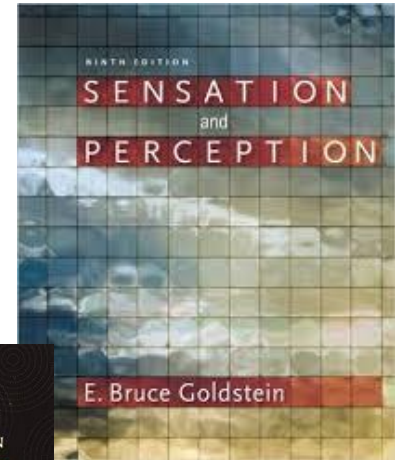
Week 2
Models, theories &
tests

Weeks 3&4
Features & objects

Week 5
Revision & Test

Your learning tools

- Text book
 - Goldstein, E. B. (2014). *Sensation and Perception*. 8th – 10th edition.
- Lectures and lecture slides
 - Blue boxes (optional)
 - Test yourself slides
 - 1 x Learning outcome
 - 3 x Multiple choice questions
- Knowledge trees



If you have questions...

- ...administrative or subject-related, please post them on blackboard in our Visual Perception folder

PSYC1071_2025

Introduction to Psychology I: Cognitive and Biological Psychology (25/26)

Content Calendar Announcements (12) Discussions Gradebook (10) Messages Analytics Groups

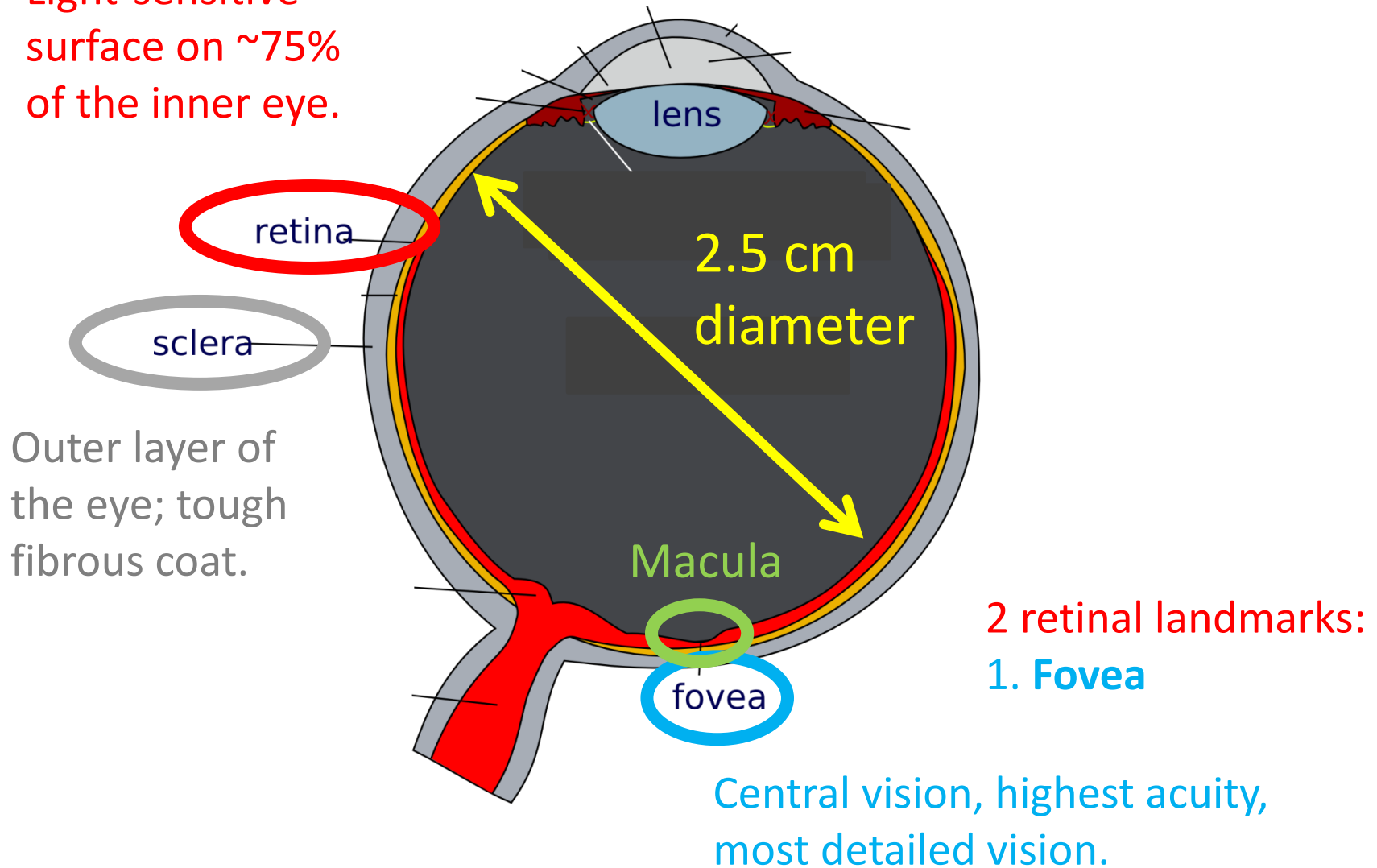
- Questions related to Visual Perception
Visible to students
- Intro 1 Biological Psychology Discussion board
Visible to students
General discussion board. Please post comments on the lectures/content/pods here.
- Questions related to Biological Psychology
Visible to students

- I will usually stick around after each lecture for a bit.

Basics of vision I: The eye

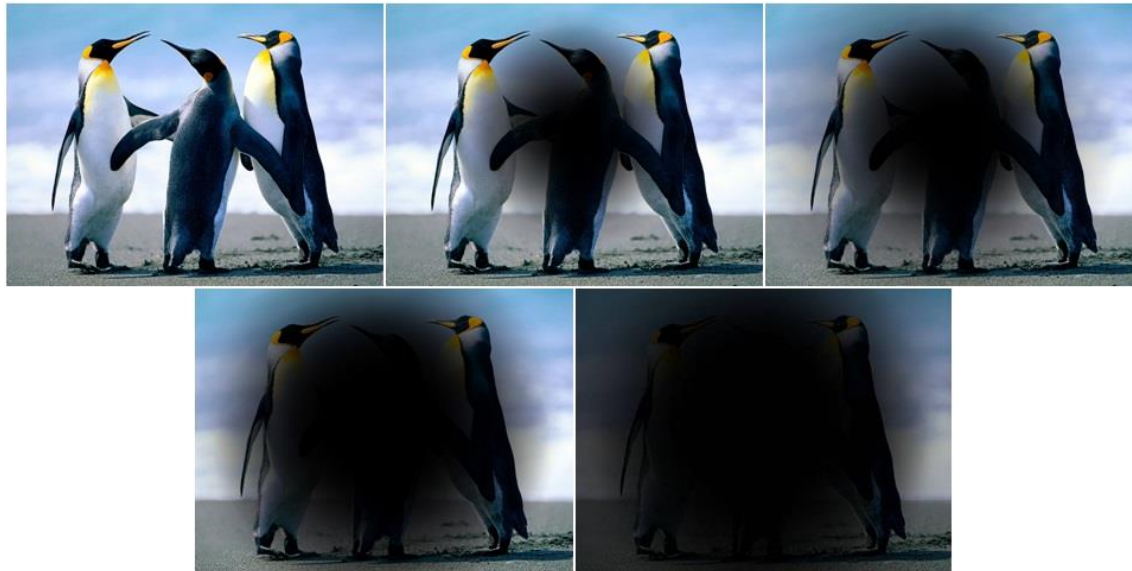
Human eye anatomy

Light-sensitive
surface on ~75%
of the inner eye.



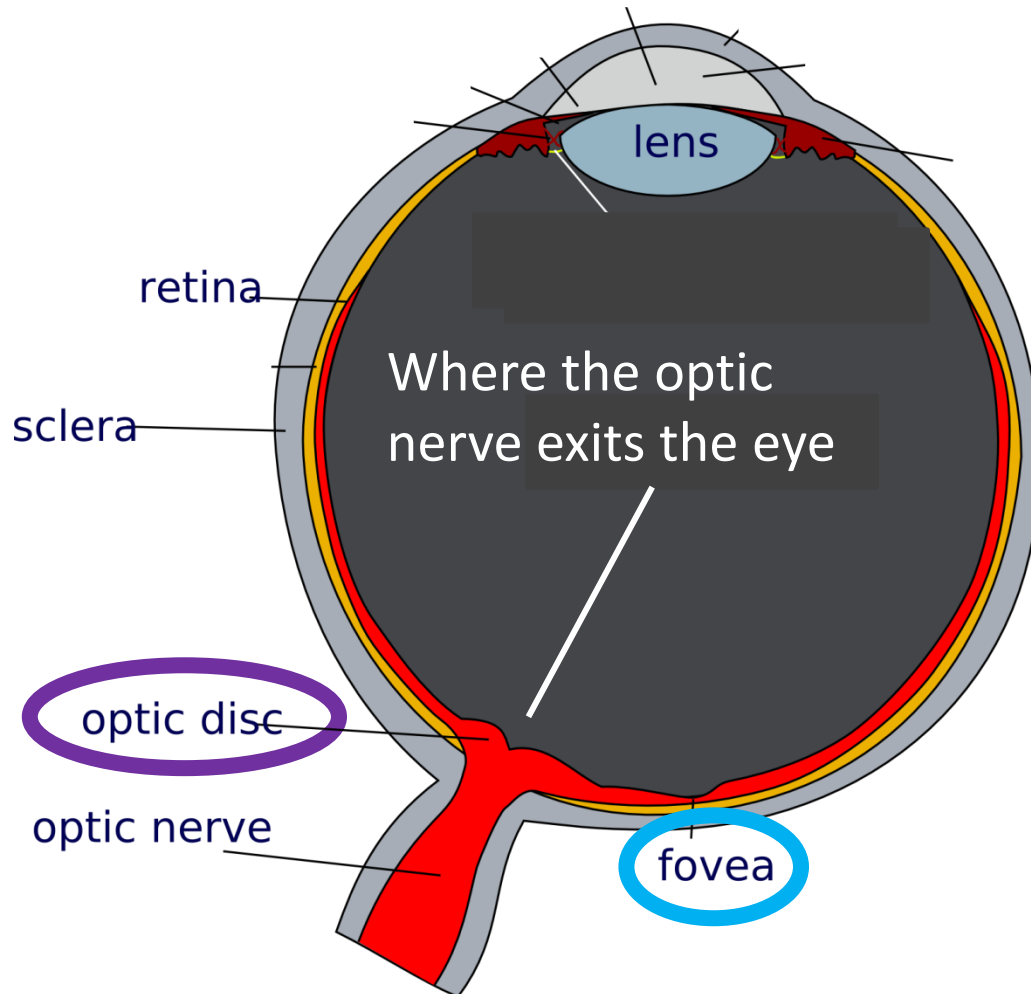
Human eye anatomy

- **Macular degeneration**
 - Deterioration of the retina in the macula (fovea).
 - **Irreversible** blindness, **loss of central vision**.



If you want to know more, look at this video about MD from the Macular Disease Foundation Australia: <https://www.youtube.com/watch?v=Q71n209PxIU>

Human eye anatomy



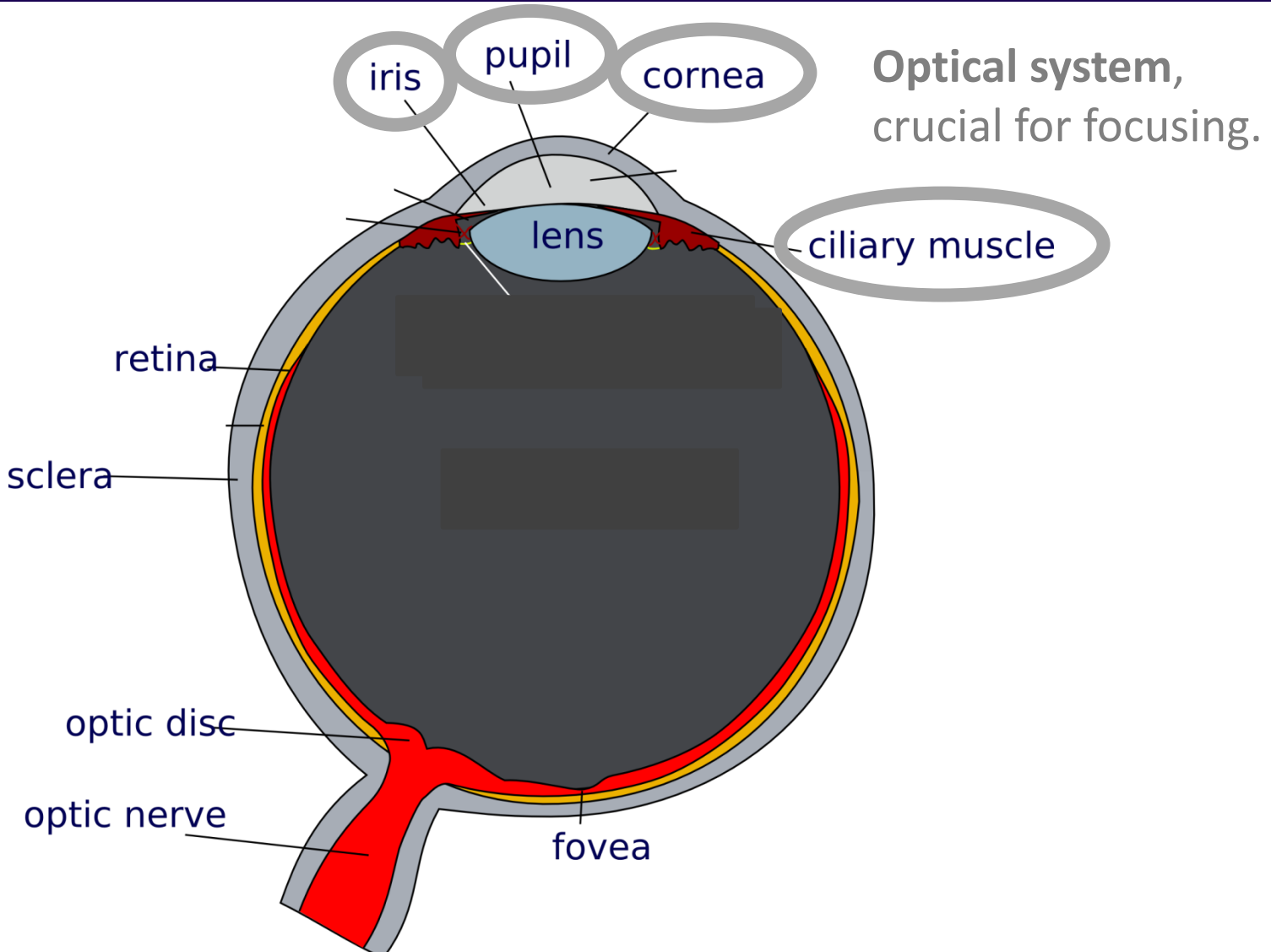
2 retinal landmarks:

1. Fovea

2. Optic disc

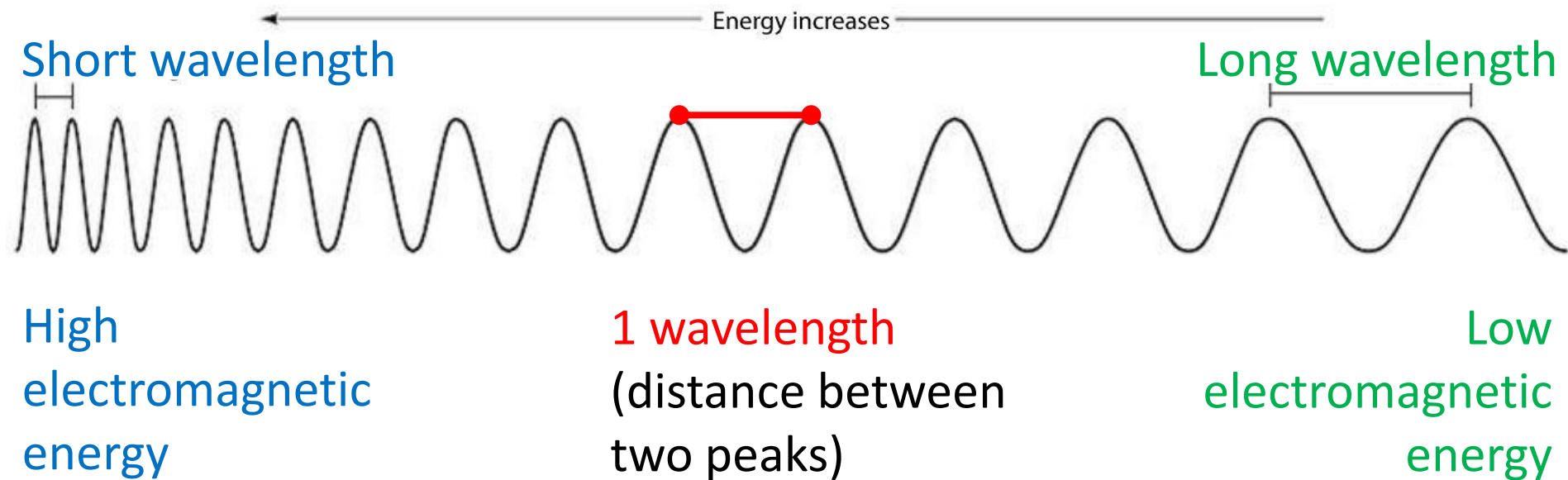
Blind spot, usually unaware, because blind spot of one eye corresponds to 'seeing' retina in the other eye.

Human eye anatomy



More **eye anatomy** here: https://www.tedmontgomery.com/the_eye/indexmacula.html

The visual process: Light

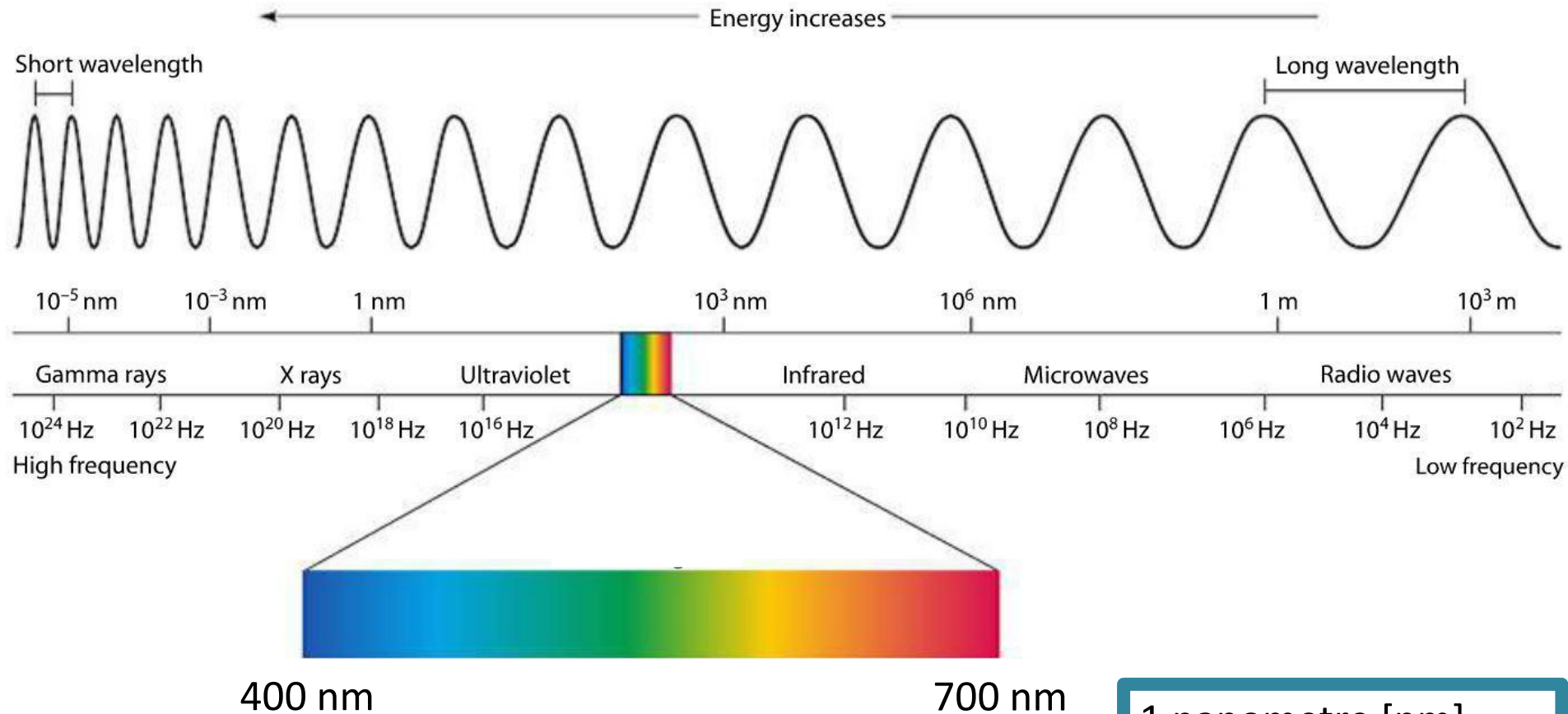


Light is **electromagnetic energy**, measured in wavelengths.

Light is the means that enables us to perceive objects.

The visual process: Light

Vision is based on **visible light**.



Spectrum of visible light

1 nanometre [nm]
= 10^{-9} meters
= **billionth** of a meter

The visual process: Transformations

- 1st transformation:

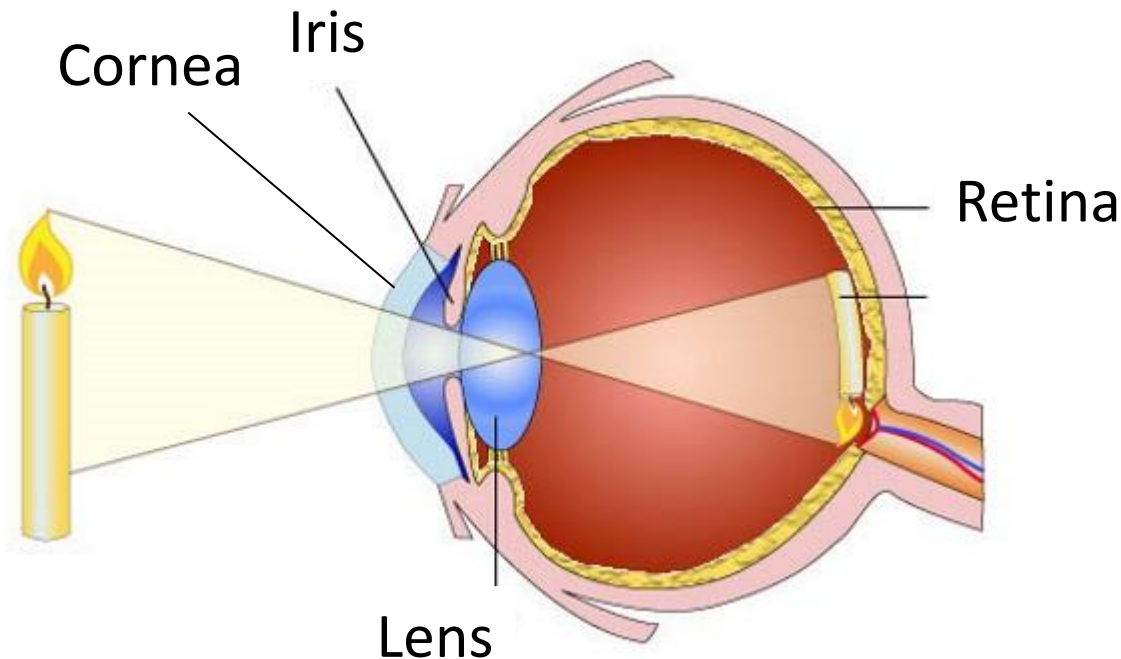
Light waves → **retinal object representation**

- 2nd transformation:

Retinal object representation → **electrical signal**

The visual process: 1st transformation

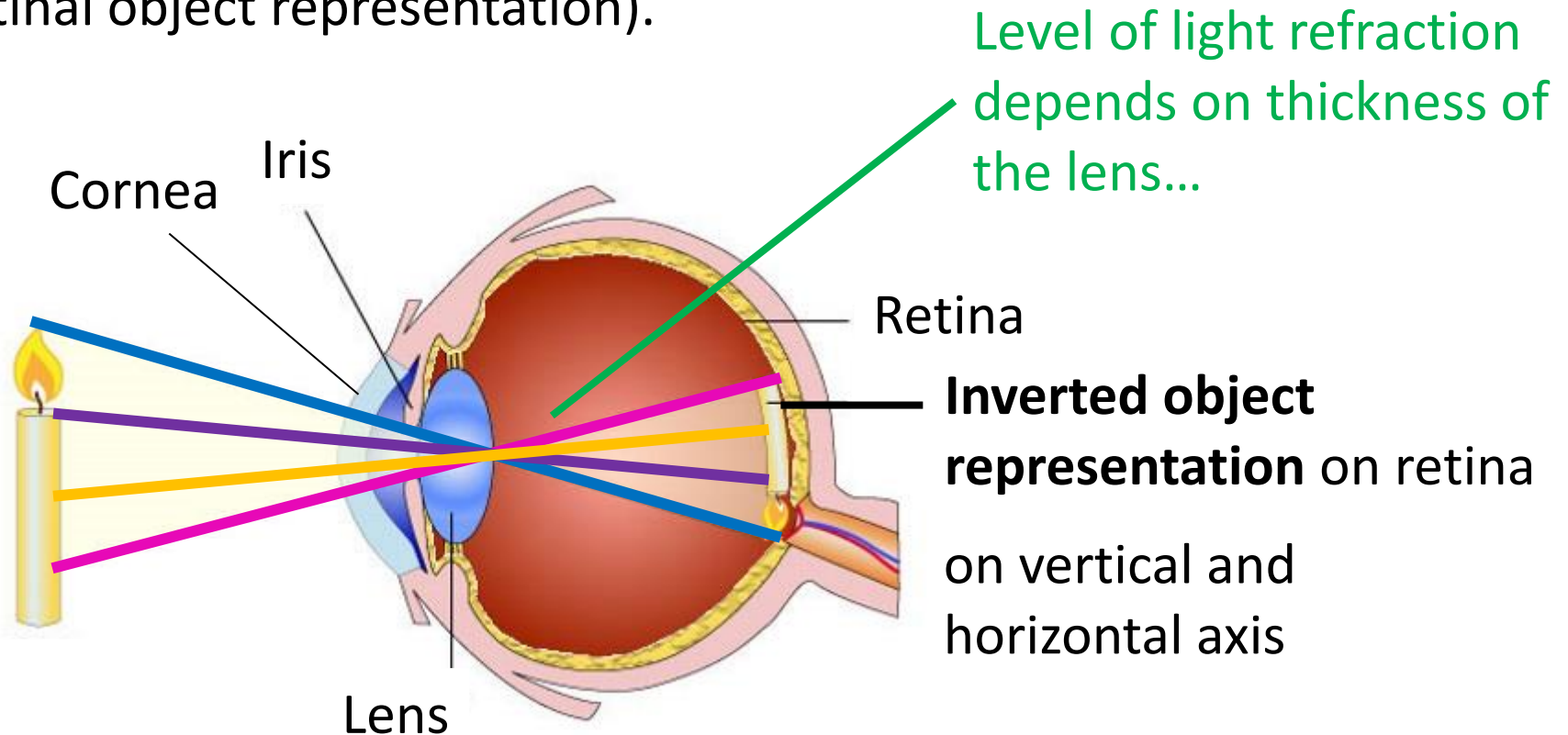
Transformation of **light** (reflected by an object) **into a retinal image** (retinal object representation).



How are the **iris and pupil** involved and what this has to do with **attractiveness**?
John Harris; *Sensation and Perception*; SAGE publications, 60-63.

The visual process: 1st transformation

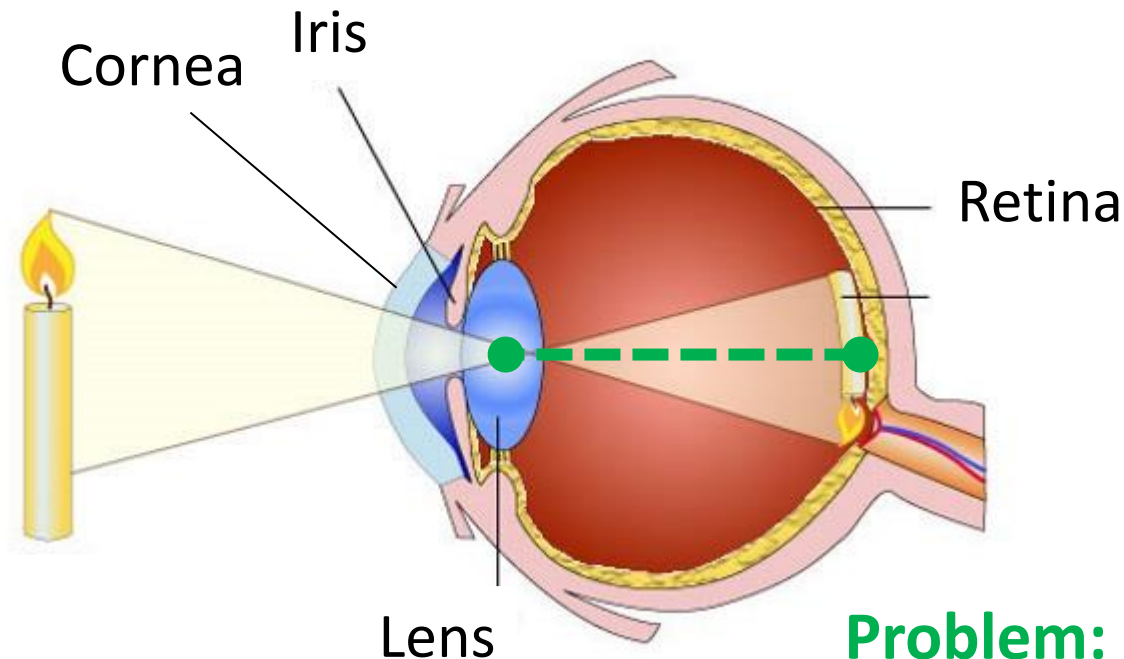
Transformation of **light** (reflected by an object) into a **retinal image** (retinal object representation).



For an excellent illustration of **inverted object representations** look at:
http://www.ivyroses.com/HumanBody/Eye/Eye_Image-Formation.php

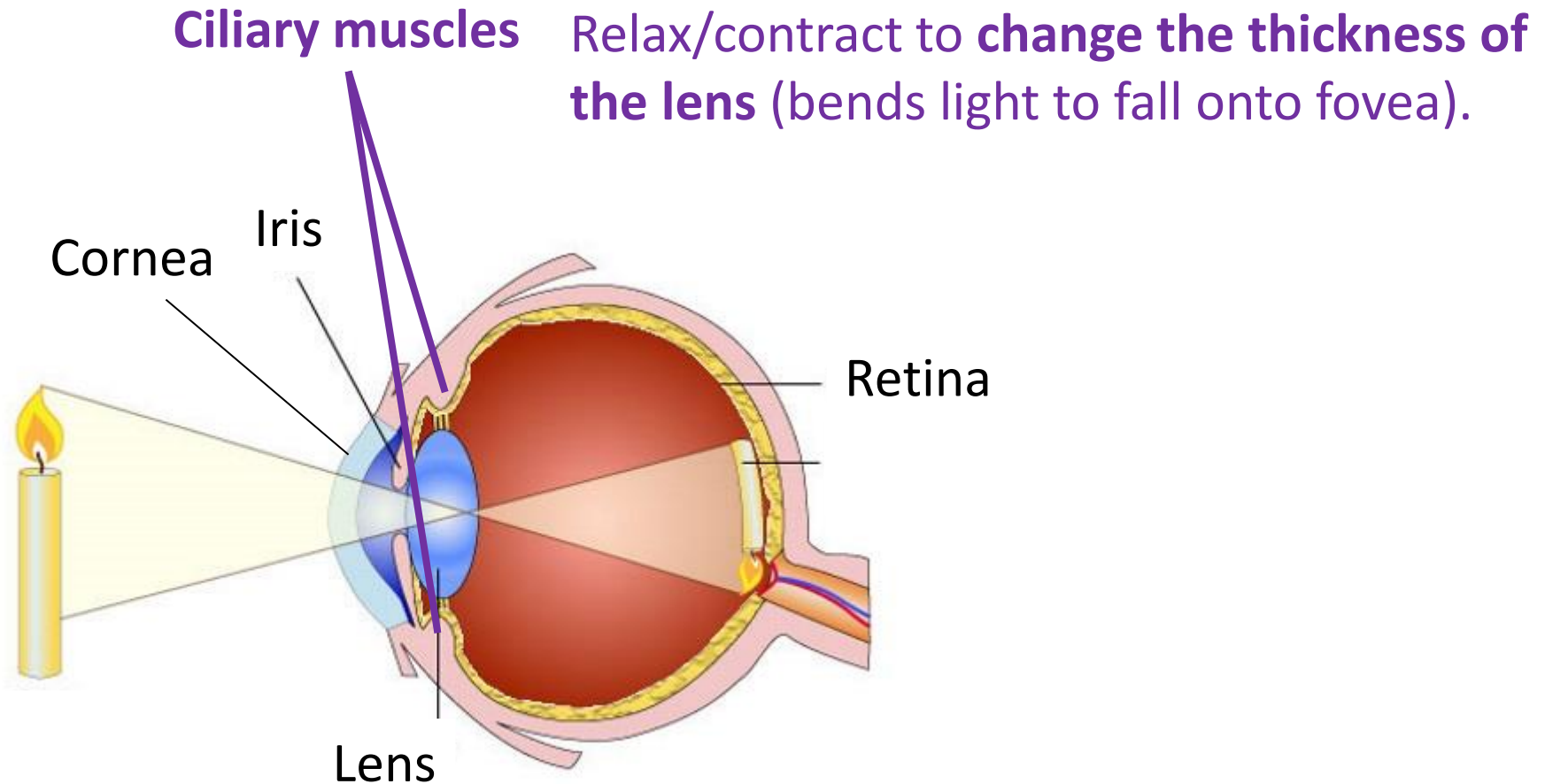
The visual process: 1st transformation

Accommodation: Focussing on near/far objects



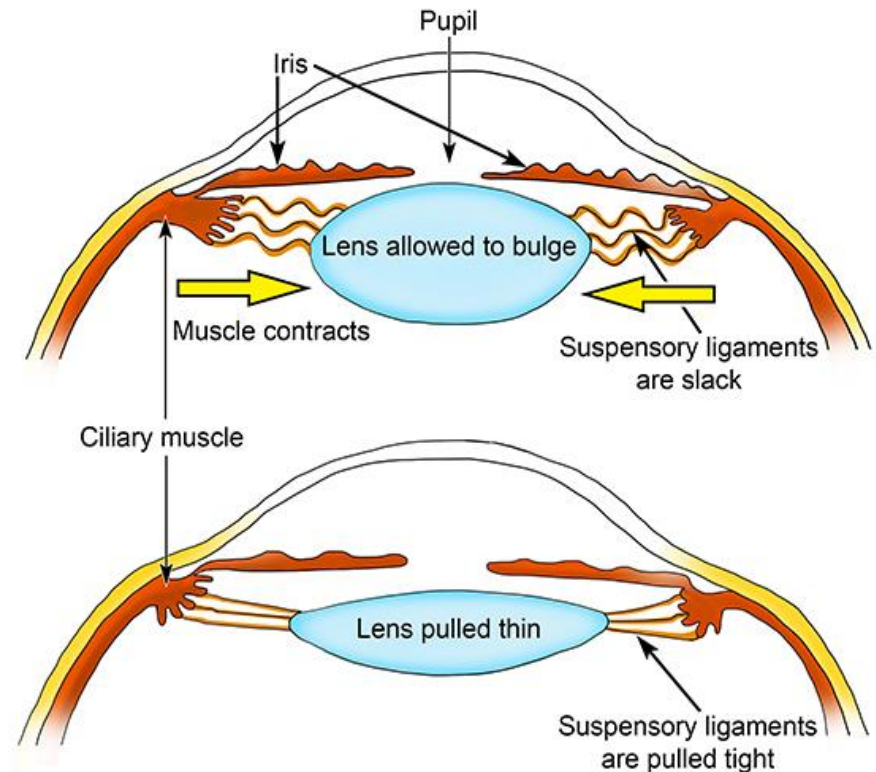
Problem:
Fixed distance between
lens and retina

The visual process: 1st transformation



The visual process: 1st transformation

Ciliary muscles Relax/contract to **change the thickness of the lens** (bends light to fall onto fovea).



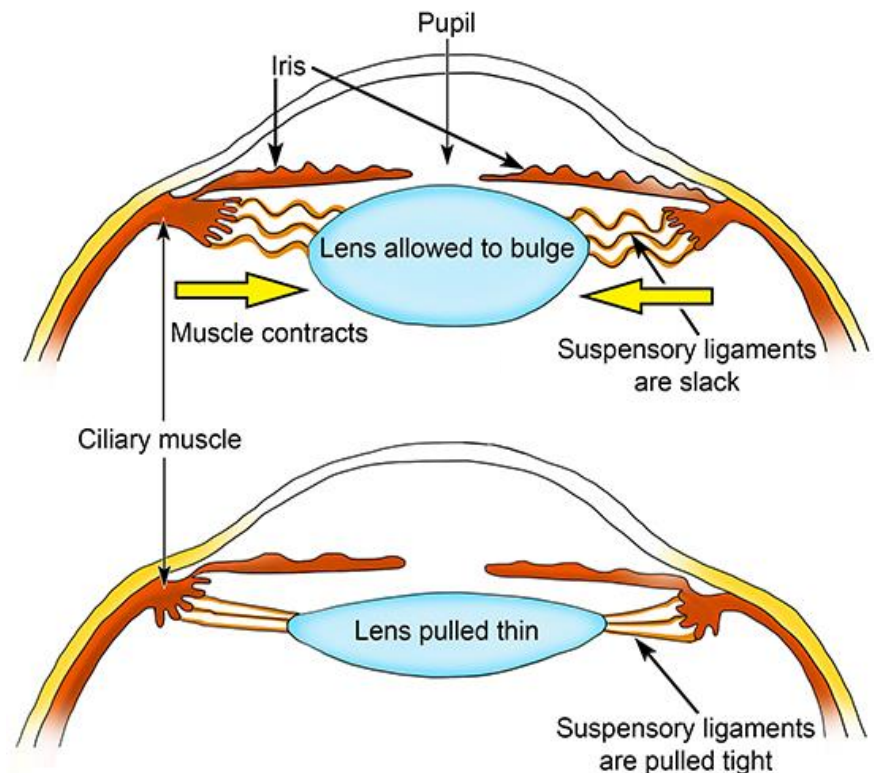
The visual process: Accommodation

- Accommodation to a **far object**

- **Relaxed** ciliary muscles

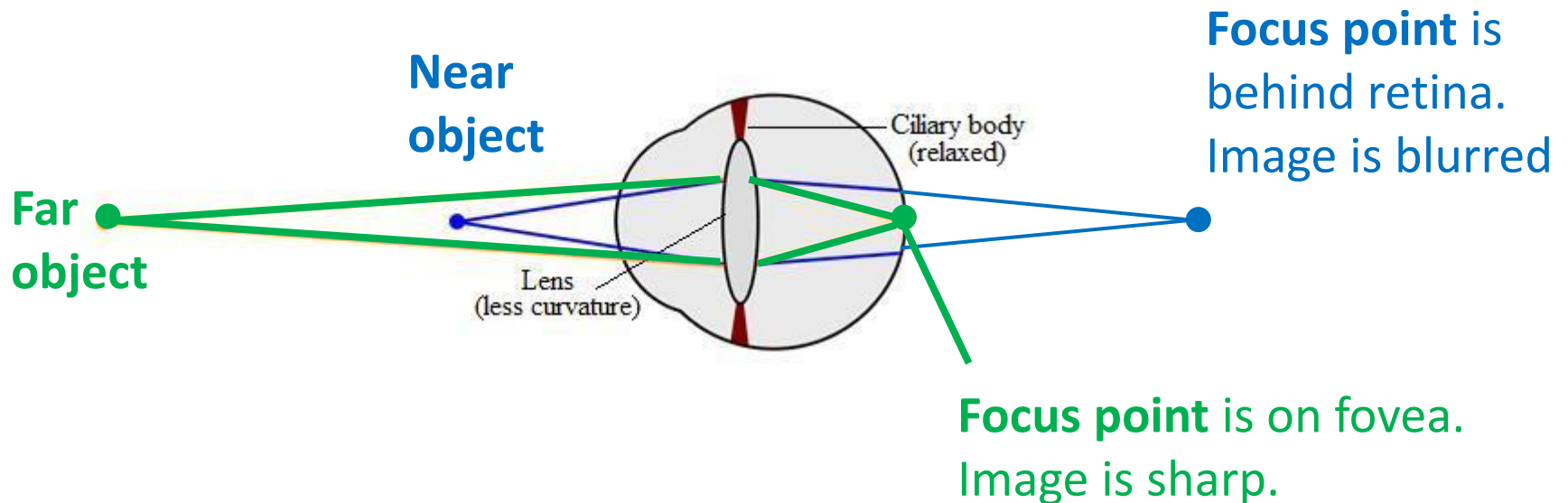
Slim lens (little curvature, little focusing power)

Light is bent only a **little**



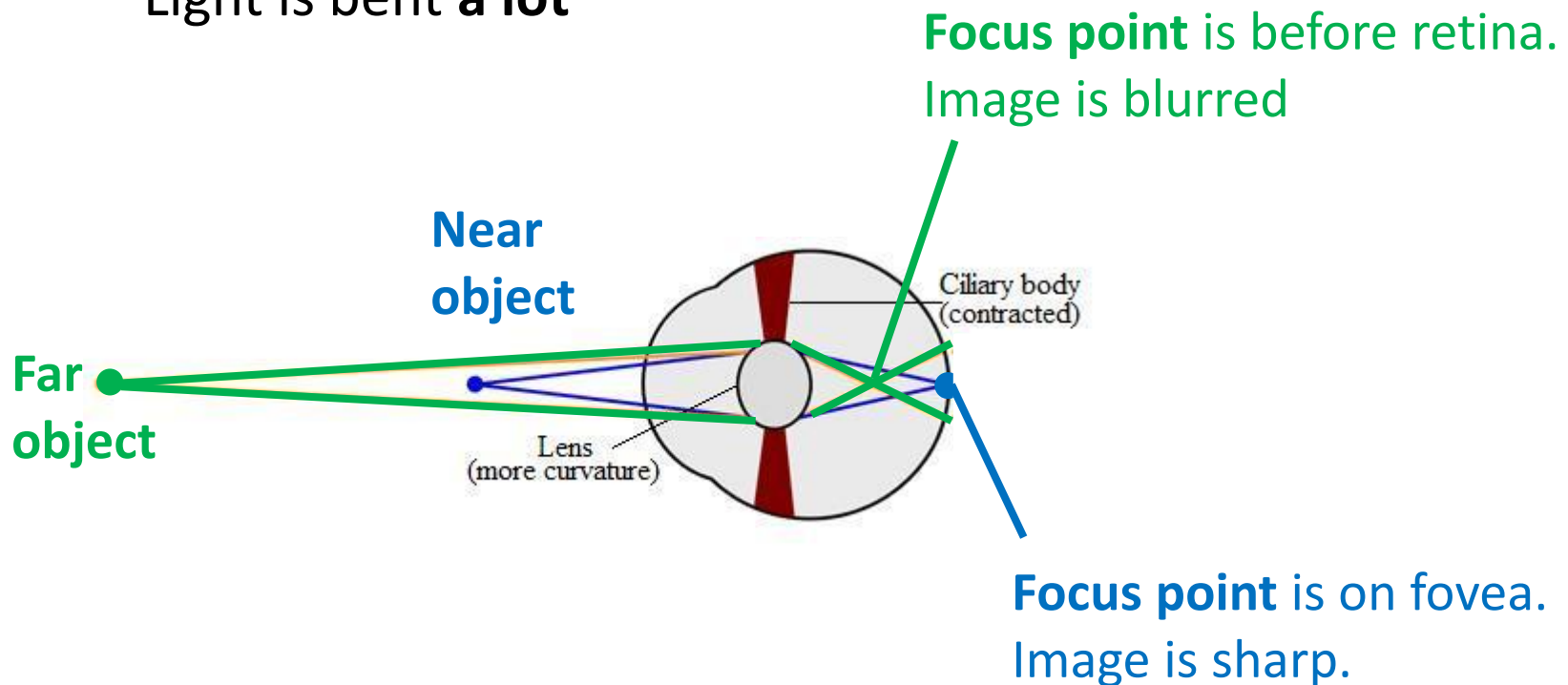
The visual process: Accommodation

- Accommodation to a **far object**
 - **Relaxed** ciliary muscles
 - Slim** lens (little curvature, little focusing power)
 - Light is bent only a **little**



The visual process: Accommodation

- Accommodation to a **near object**
 - **Contracted** ciliary muscles
 - Thick** lens (strong curvature, strong focusing power)
 - Light is bent **a lot**



The visual process: Accommodation

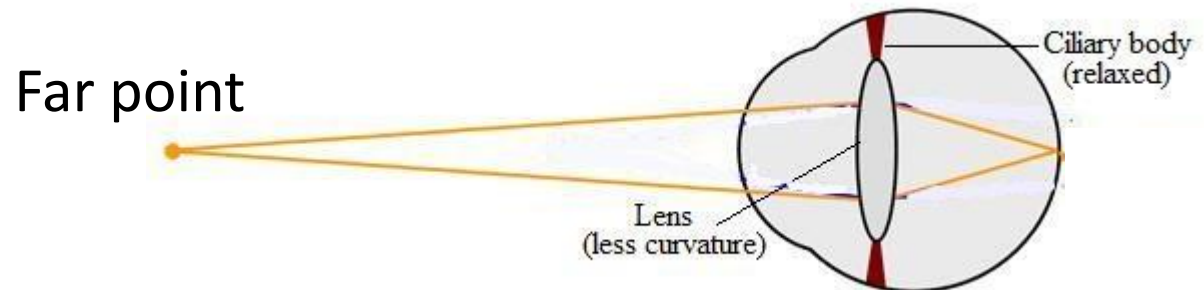
- Some vocabulary

- **Accommodation**

Process by which the eye changes optical power to focus on an object as its distance varies.

- **Far point**

The maximum distance of an object from the eye for which a clear image of the object can be seen.



The visual process: Accommodation

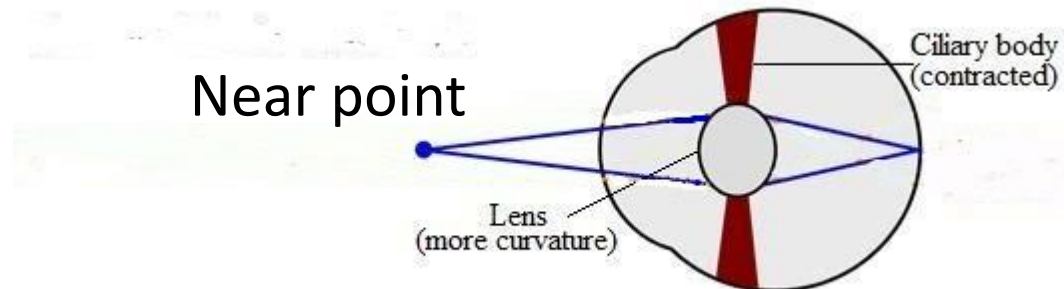
- Some vocabulary

- **Near point**

The minimum distance of an object from the eye for which a clear image of the object can be seen.

Accommodation limit.

Increases with increasing age.



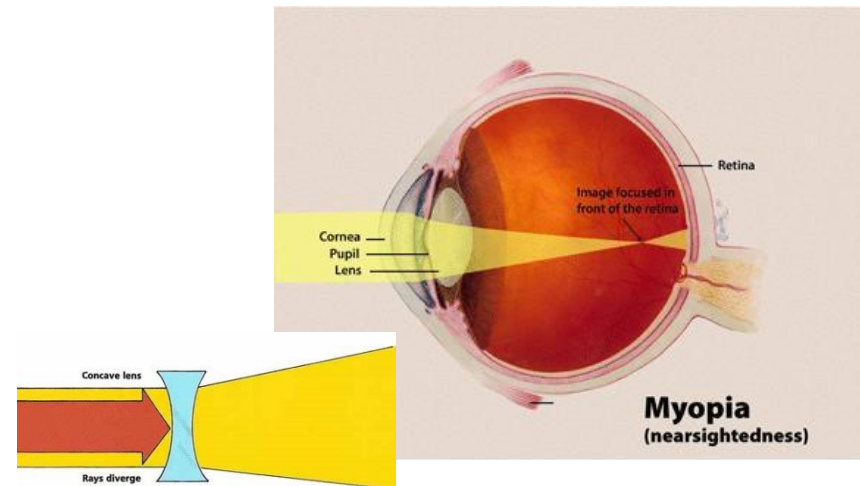
The visual process: Accommodation

- Two common visual problems
 - **Myopia** (nearsightedness)

Far objects are out of focus, because the lens is too thick (bends the light too much) or the eyeball is too long.

Requires **concave correction** lenses.

Concave correction to **diverge** light before it enters the eye.



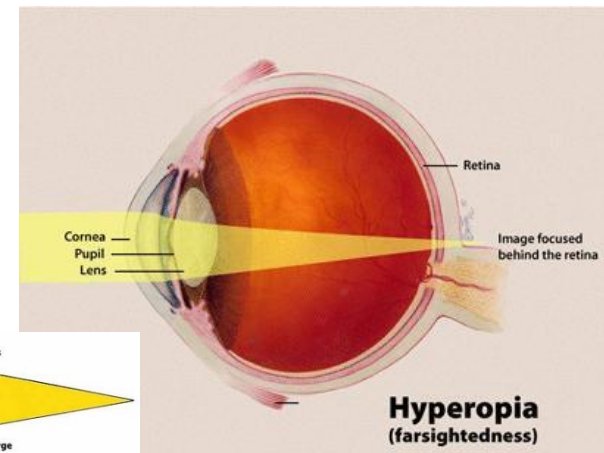
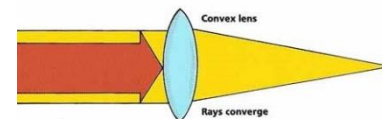
The visual process: Accommodation

- Two common visual problems
 - **Hyperopia** (farsightedness)

Near objects are out of focus, usually because the eyeball is too short (lens cannot get any thicker).

Requires **convex correction** lenses.

Convex correction to **converge** light before it enters the eye.



The visual process: Transformations

- 1st transformation:

Light waves → **retinal object representation**

- 2nd transformation:

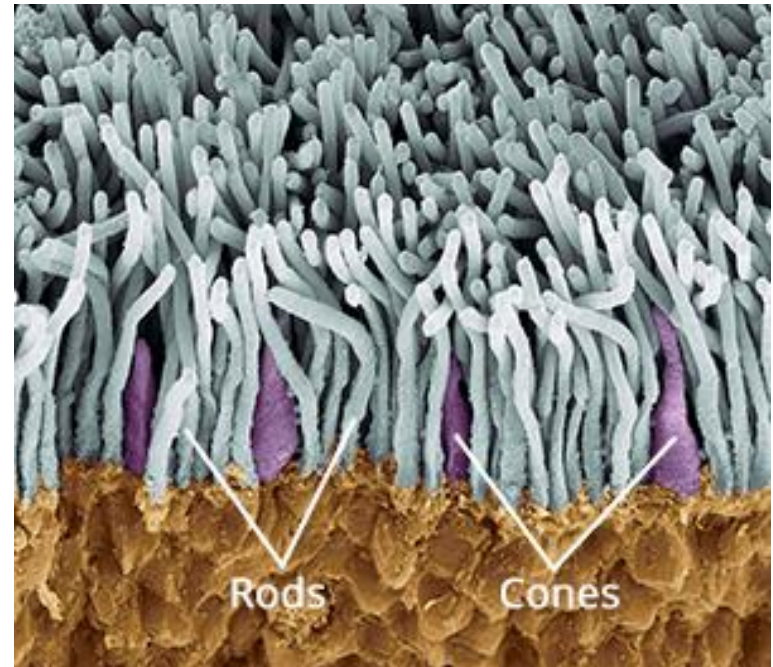
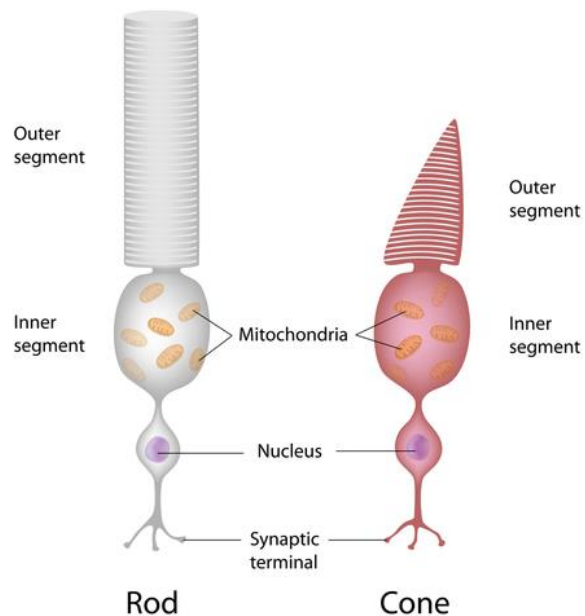
Retinal object representation → **electrical signal**

The visual process: 2nd transformation

Transduction

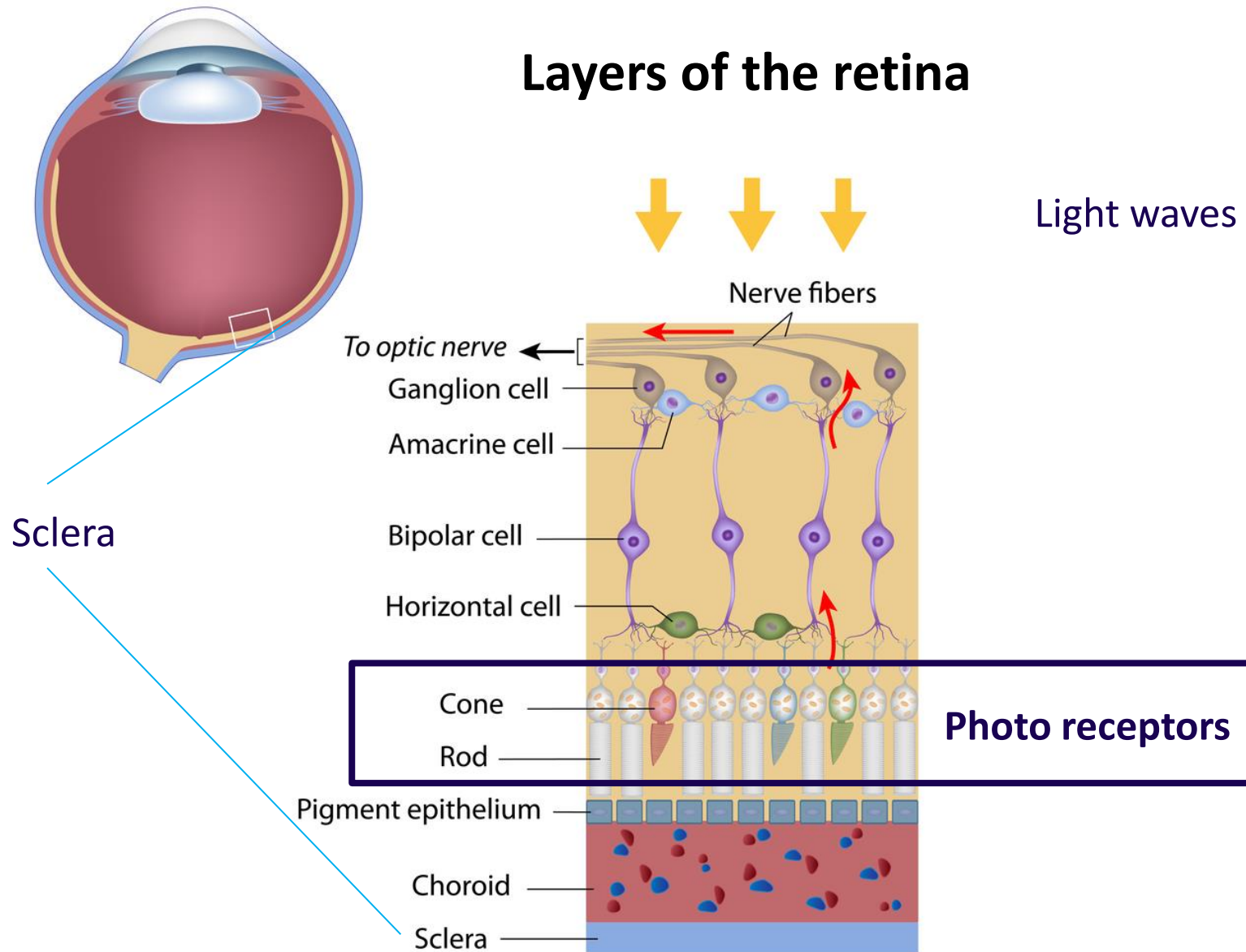
The retinal image is transformed into electrical energy.

Photo receptors

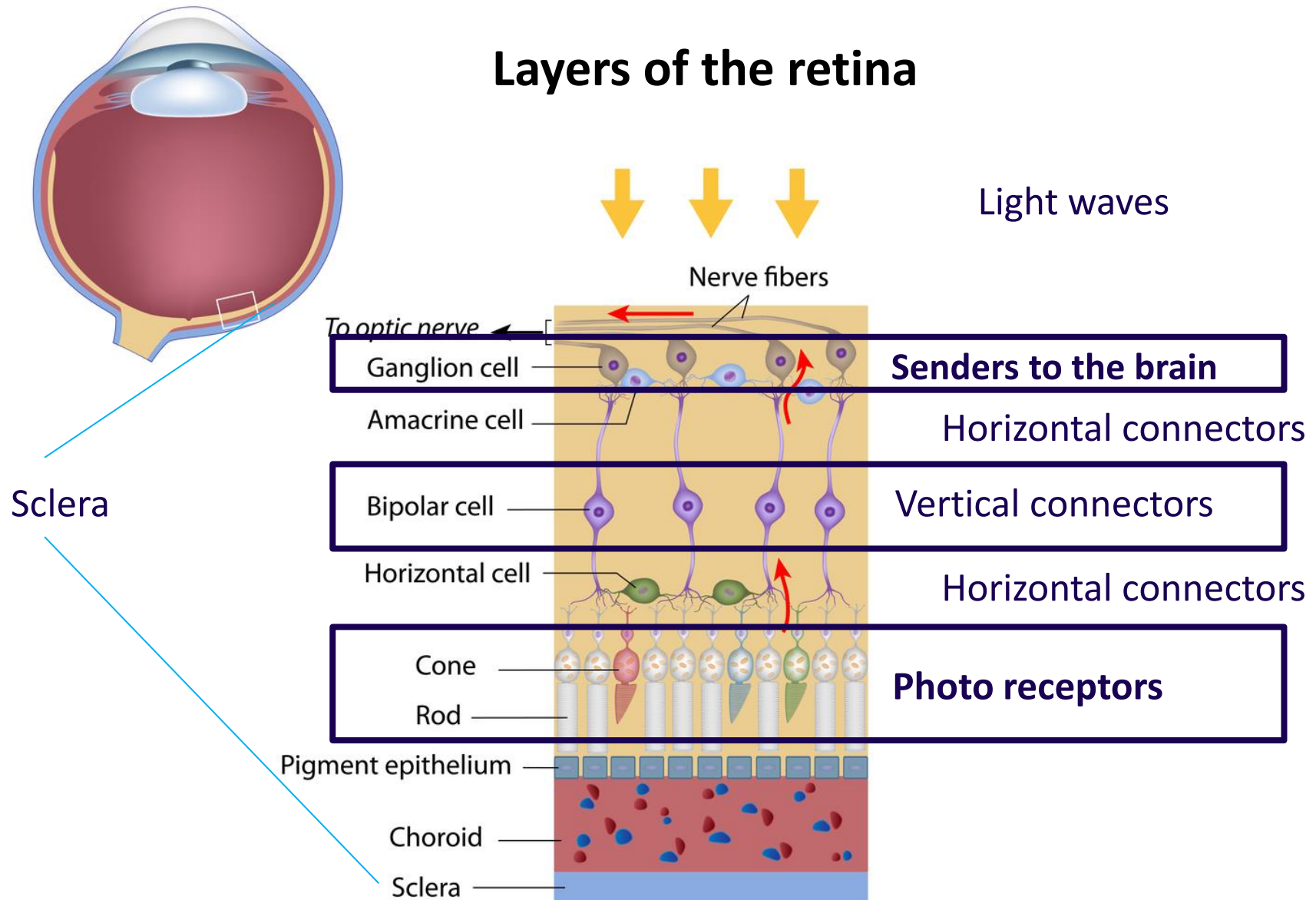


Coloured scanning electron micrograph
– 1,800 times magnification.

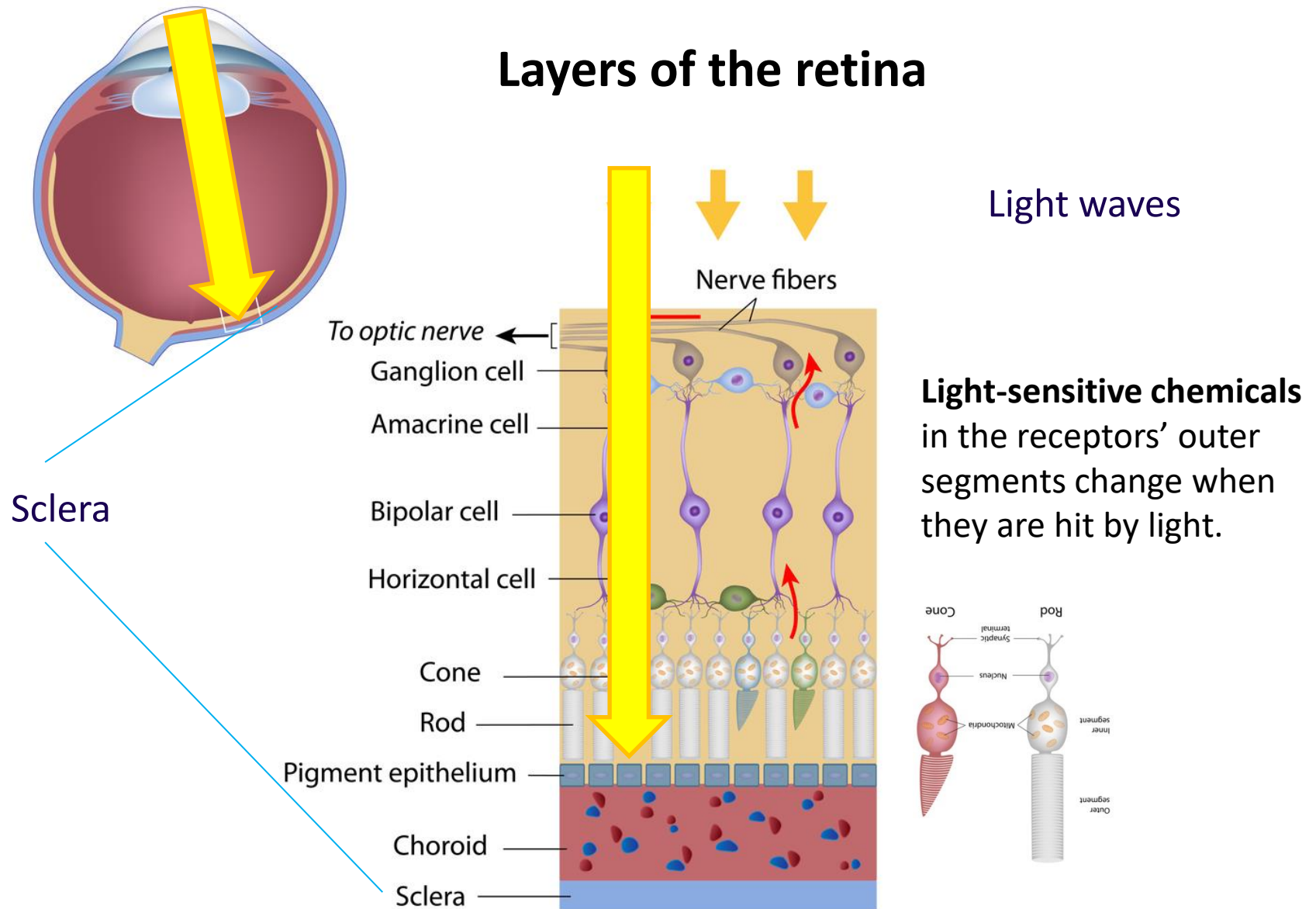
The visual process: 2nd transformation



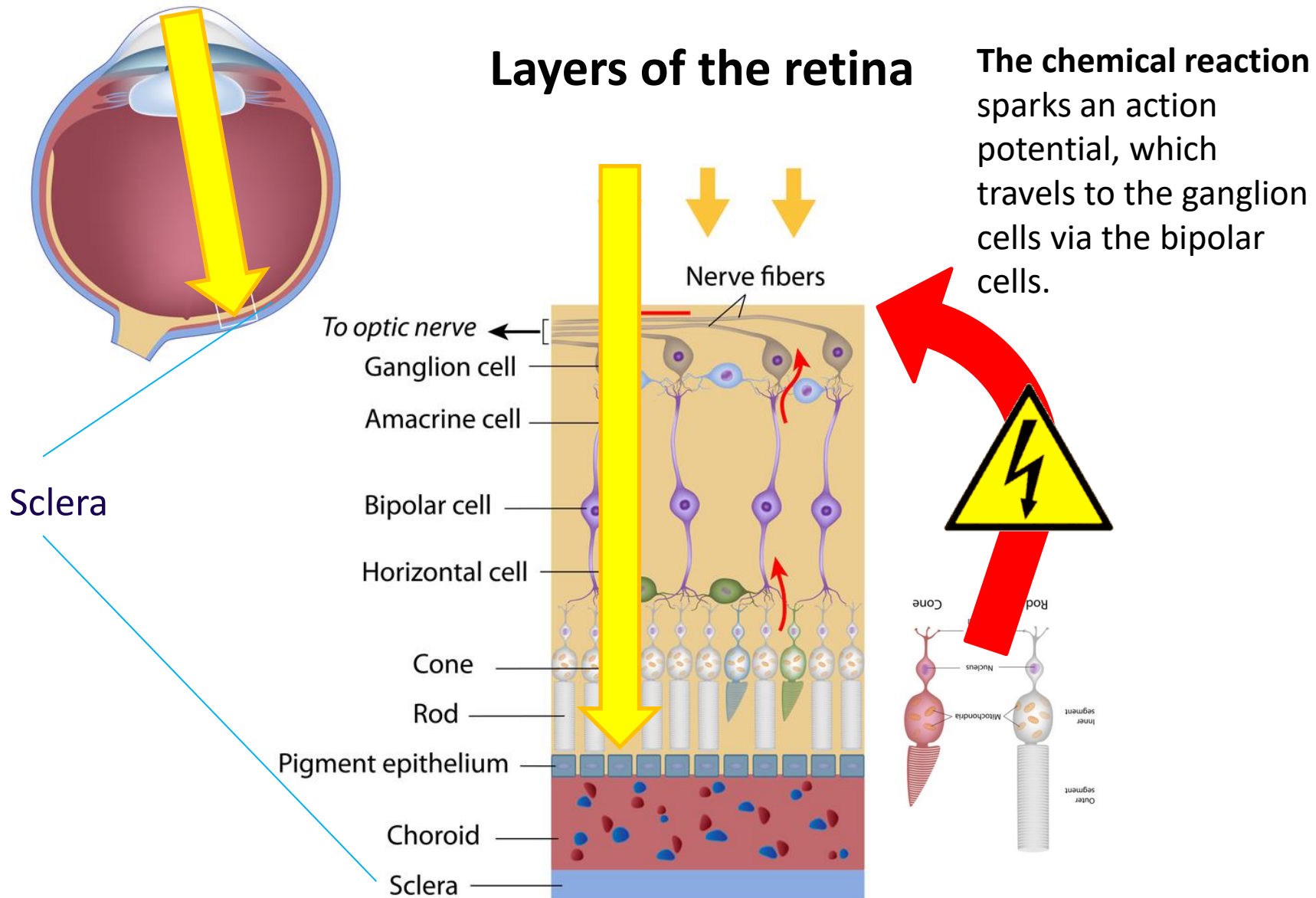
The visual process: 2nd transformation



The visual process: 2nd transformation



The visual process: 2nd transformation



The visual process: 2nd transformation

Optic disc (blind spot):

No photo receptors at that retinal location.

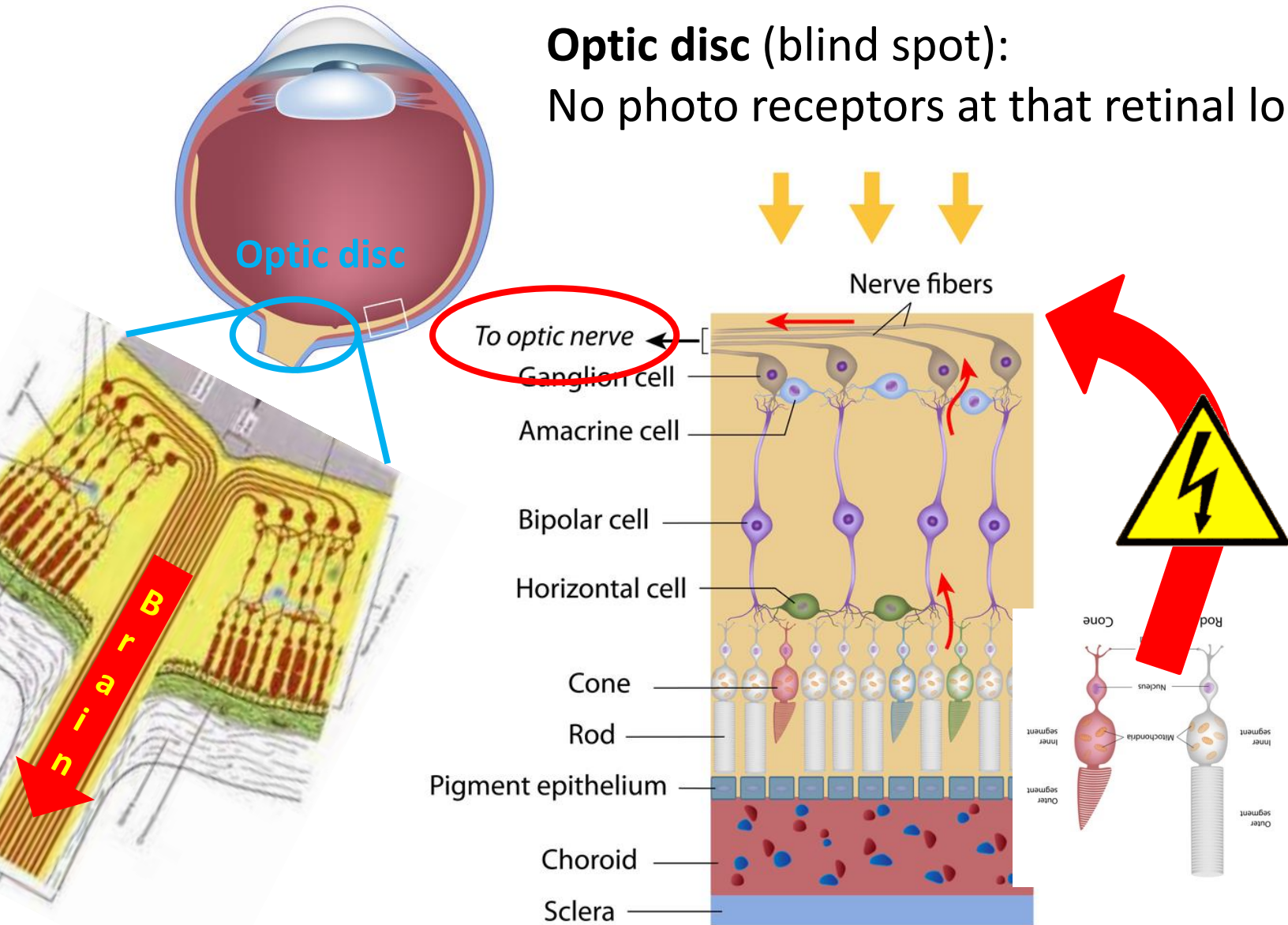


Photo receptors

- Rods and cones differ in:
 1. Absolute number
 2. Density and distribution in the retina
 3. Ability to dark adaptation
 4. Absolute sensitivity
 5. Acuity
 6. Spectral sensitivity
 7. Colour vision

Can you come up with a **mnemonic** for the 7 differences?
If so, **please share** it with me!

Photo receptors

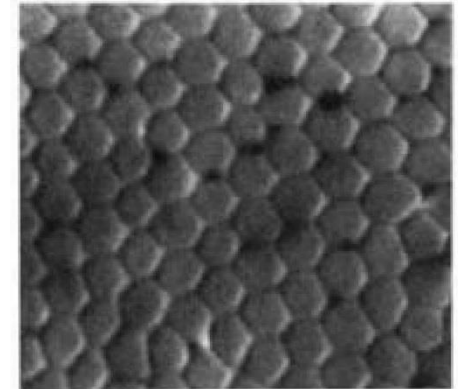
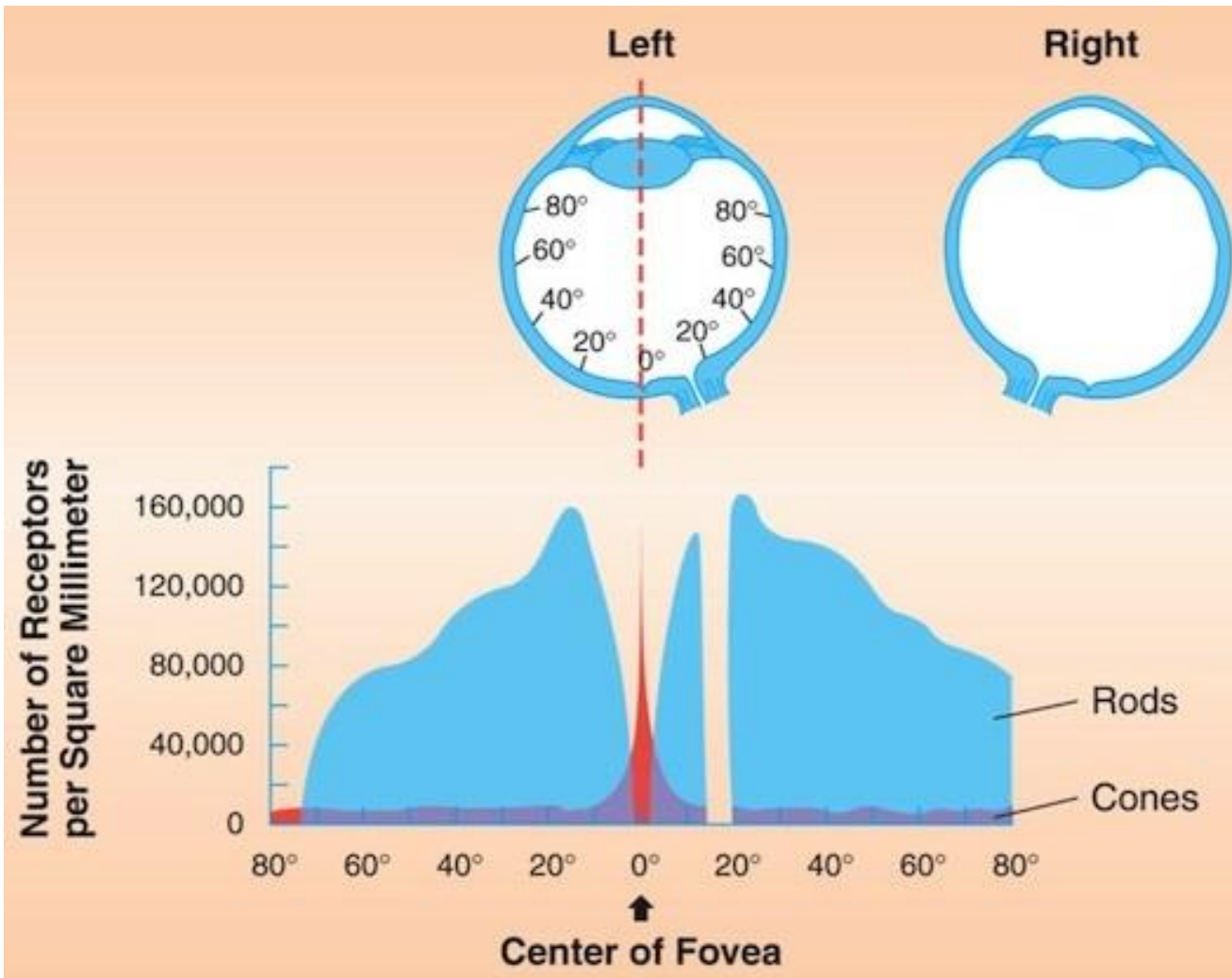
1. Absolute number

- About 120 million rods...
- About 6 million cones...

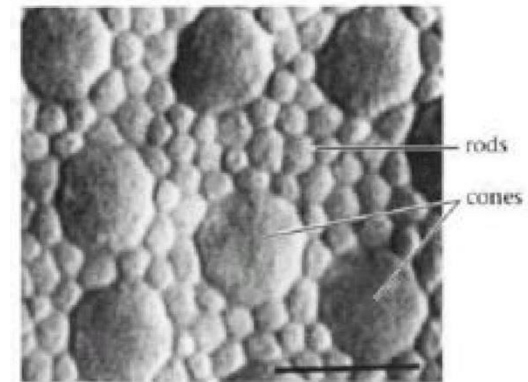
...in each human eye.

Photo receptors

2. Density and distribution in retina



Cones in Fovea



Cones and rods in periphery

Photo receptors

3. Dark adaptation (increased sensitivity to light)

Sensitivity to light **LOW**

(**Day**: a lot of light
needed for sensation)

(**Night**: little light
needed for sensation)

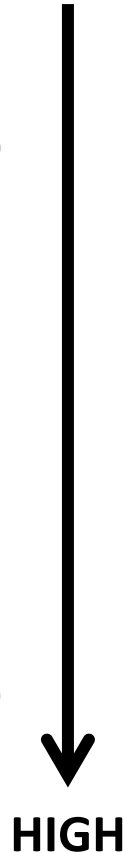


Photo receptors

3. Dark adaptation (increased sensitivity to light)

Sensitivity to light

LOW

(**Day**: a lot of light needed for sensation)

(**Night**: little light needed for sensation)

HIGH

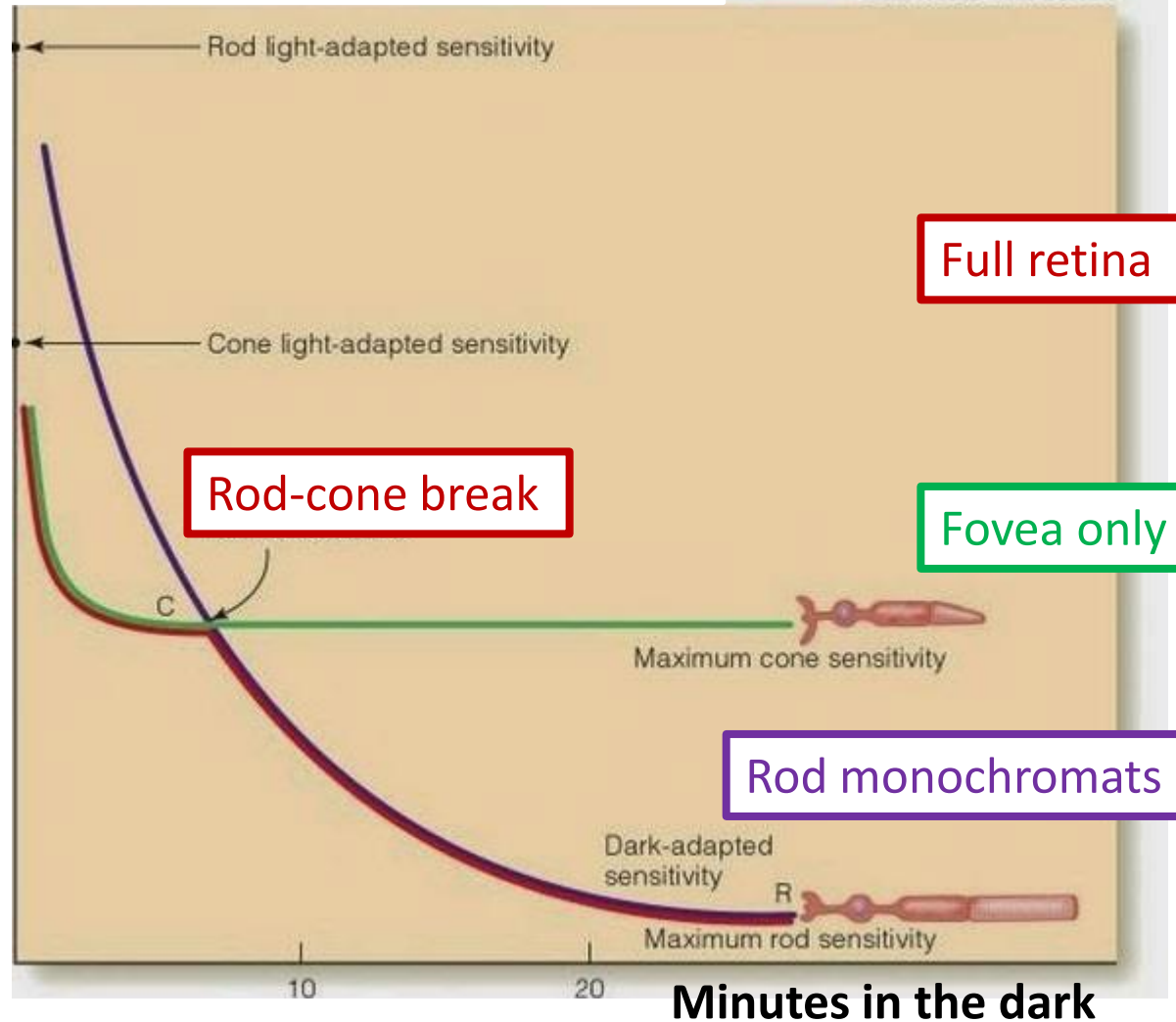


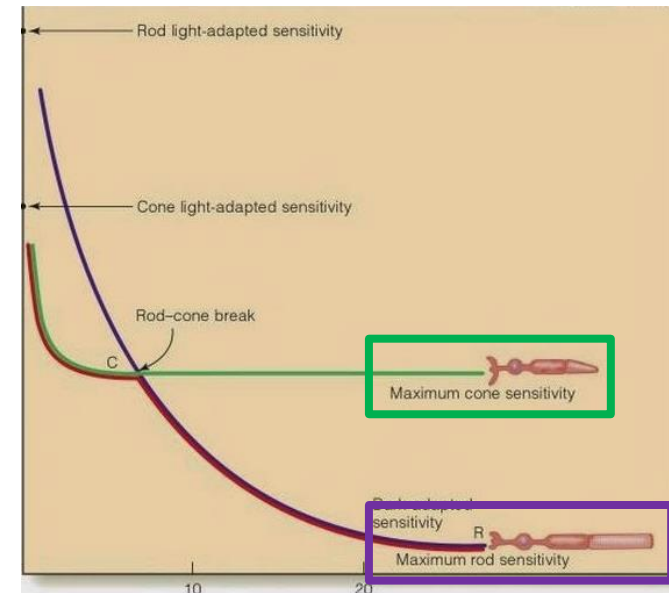
Photo receptors

4. Absolute light sensitivity

- Is **higher in rods** than cones

5. Visual acuity

- Is **better with cone** than rod vision
i.e., in light-adapted eye



→ Both have to do with **neural convergence**.

Neural convergence

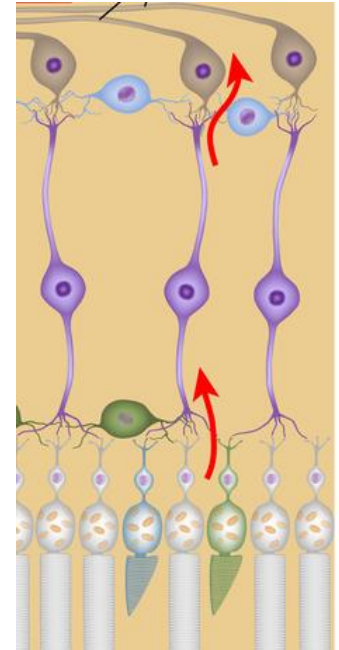
- **Many** neurons synapse onto **fewer** neurons.



120 mil rods
6 mil cones



1 mil ganglion cells



- On average, 1 ganglion receives input from:
120 rods and 6 cones
- i.e., there is **more convergence for rods than cones**.

More convergence for rods than cones

4. Increased light sensitivity in rods (ability to detect light)

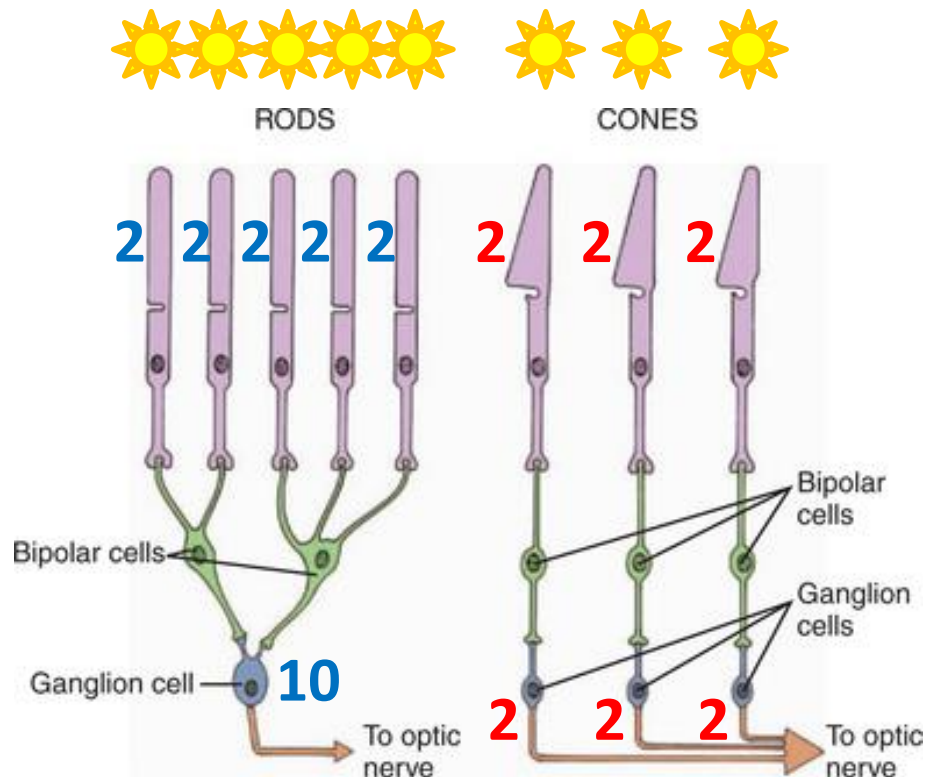
Assumptions:

- Any receptor that detects a light fires at a rate of 2.
- Firing rates sum up during convergence.

A ganglion cell that receives input from **1 cone** will fire at a rate of 2.

A ganglion cell that receives input from **5 rods** will fire at a rate of 10.

i.e., higher likelihood that light is detected by a group of rods than cones.

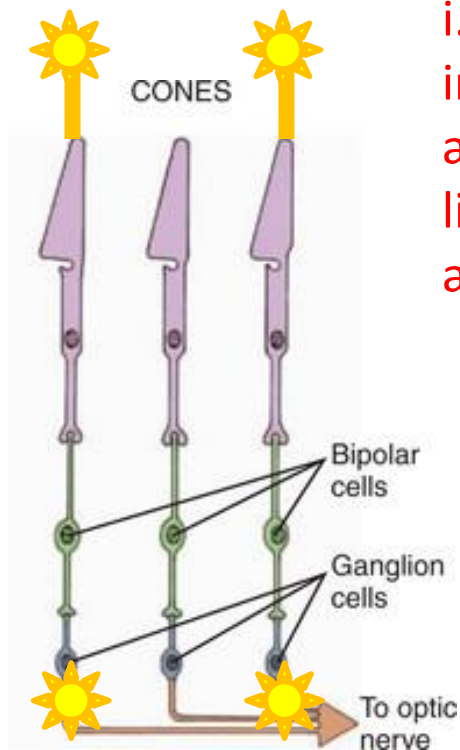


Less convergence for cones than rods

5. Increased acuity in cones (detail vision)

Separate ganglion cells receive input from separate cones,

i.e., detailed information about different light sources are available.



Same ganglion cell receives input from multiple rods,

i.e., information about different light sources is combines.

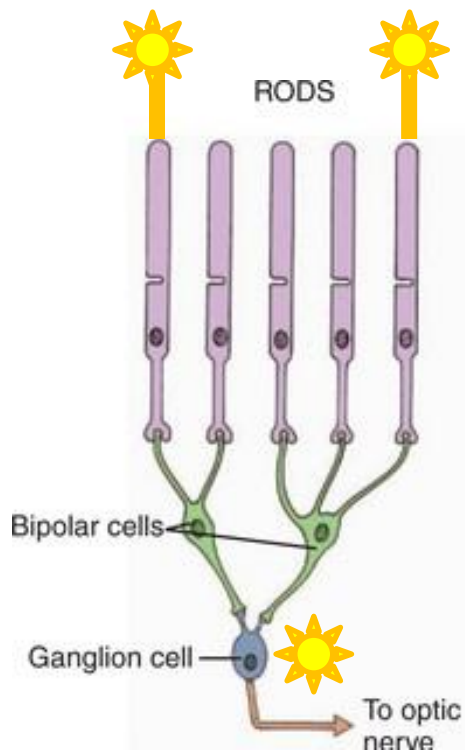


Photo receptors

6. Spectral sensitivity

- **Rods** are more sensitive to **shorter wavelengths**, maximum at 500 nm.
- **Cones** are more sensitive to **longer wavelengths**, maximum at 560 nm.

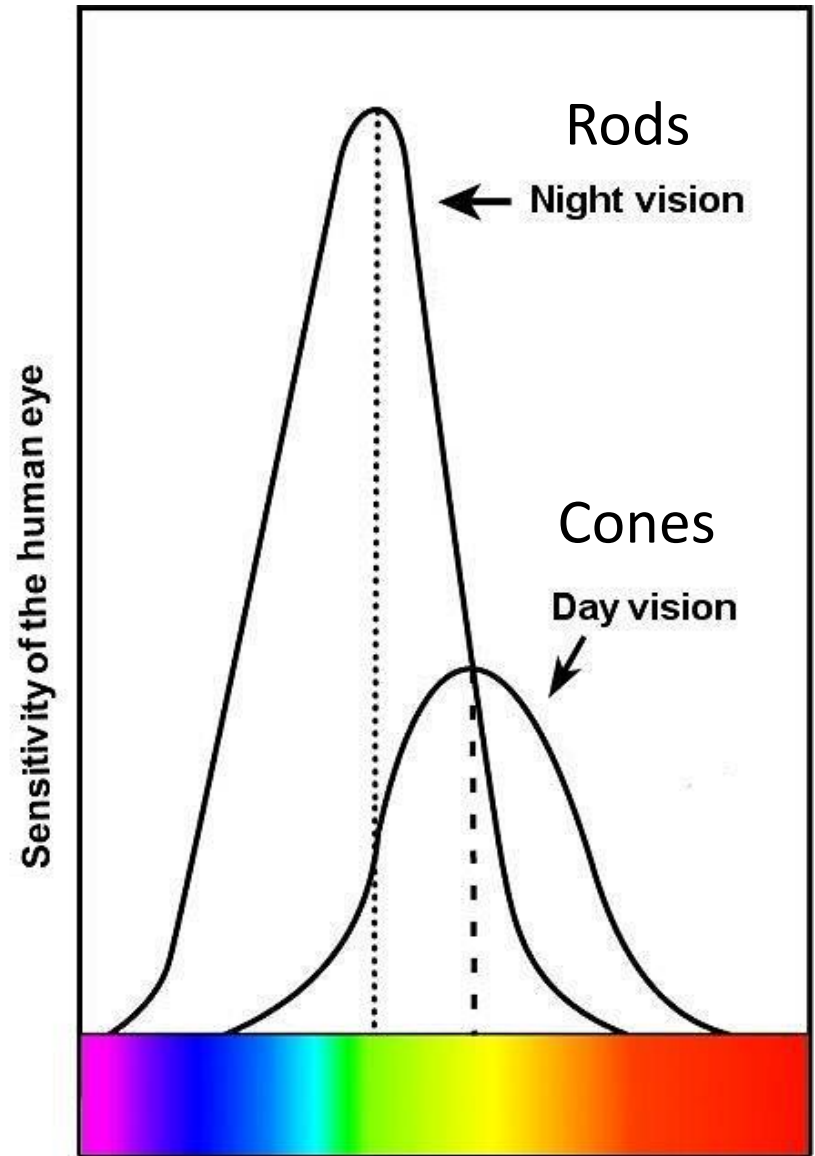


Photo receptors

6. Spectral sensitivity

– Purkinje shift

Increased sensitivity to short wavelengths in dark-adapted eye.

(Johann Purkinje, 1825)

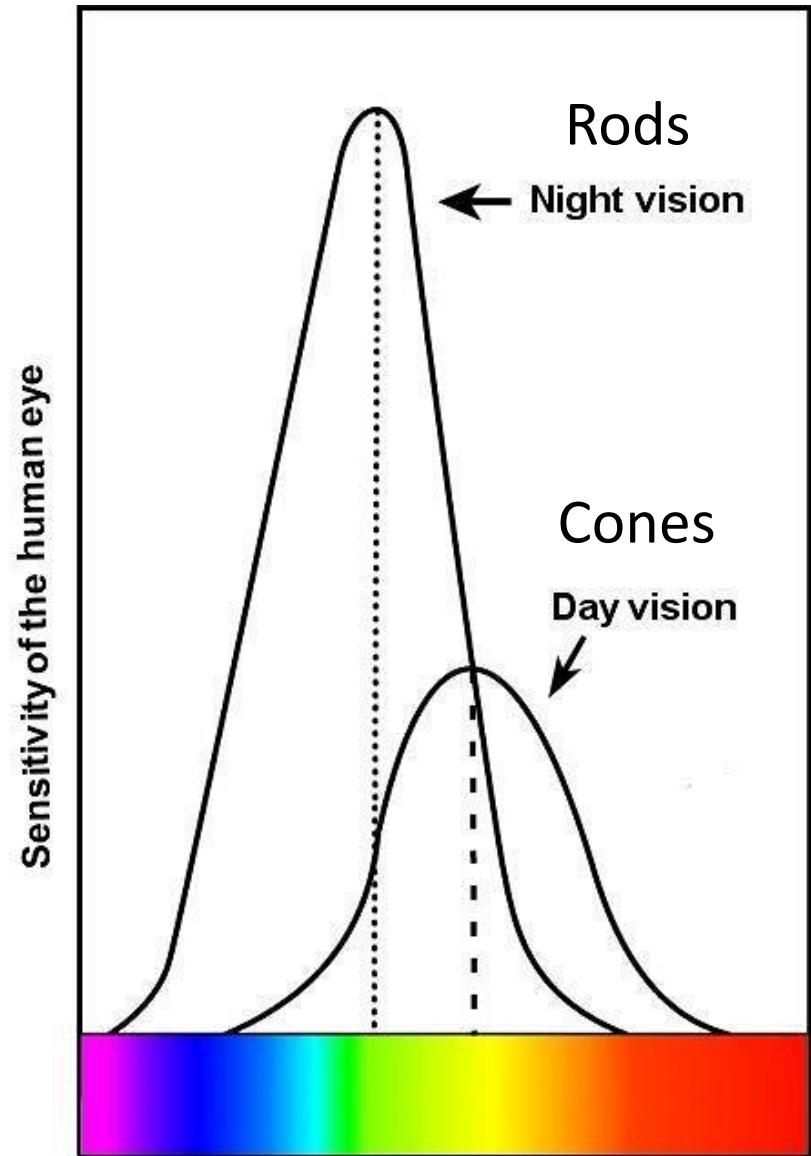


Photo receptors

6. Spectral sensitivity

— Purkinje shift

Increased sensitivity to short wavelengths in dark-adapted eye.

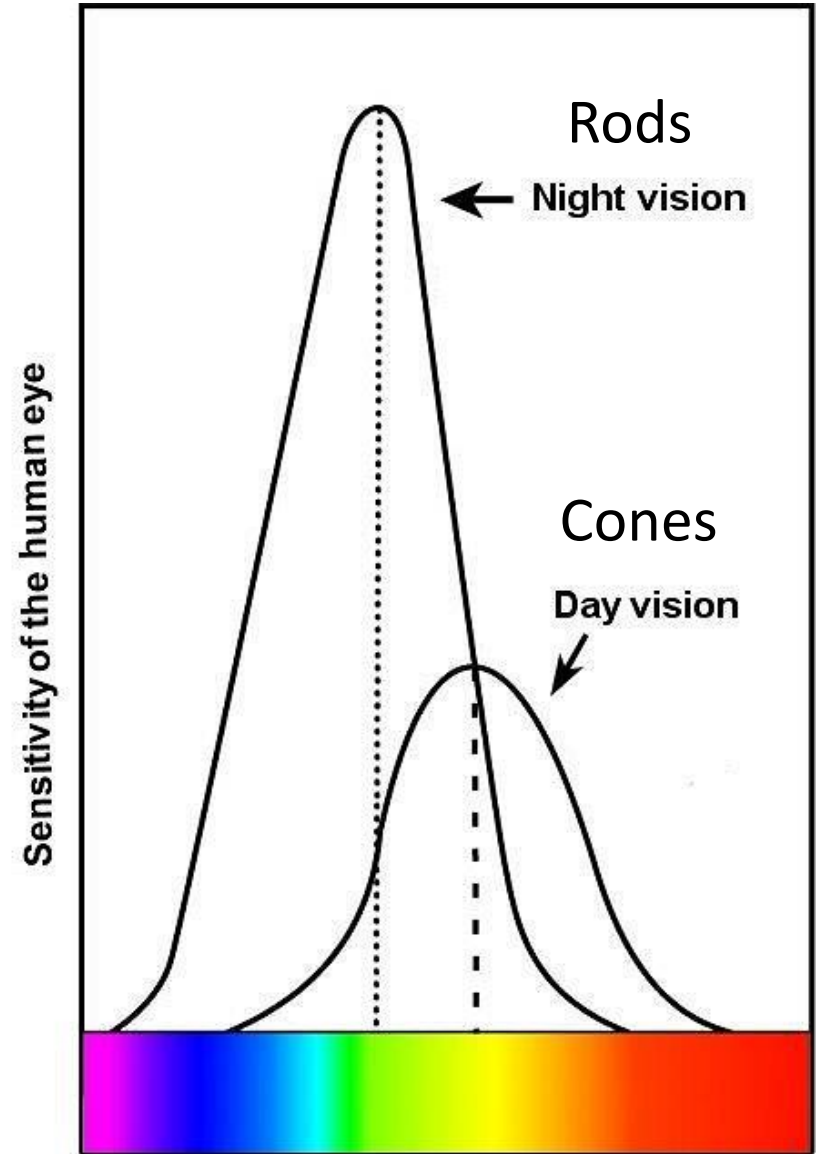
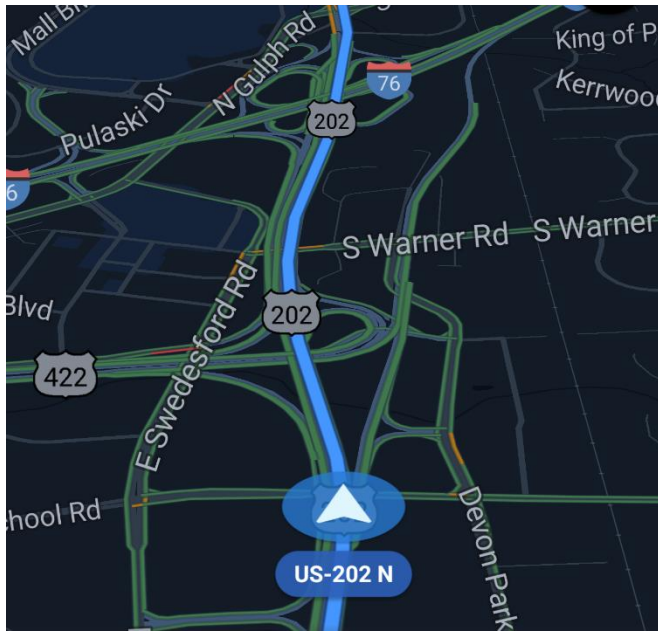


Photo receptors

7. Colour vision

- Rod vision = dark-adapted vision
operates at low luminance
No colour sensation
- Cone vision = light-adapted vision
operates at high luminance
S cones (respond to short wavelengths, blue)
M cones (respond to medium wavelengths, green)
L cones (respond to long wavelengths, red)



Two types of vision

- **Photopic vision**

Cone-dominated, foveal and peripheral vision, light-adapted vision (insensitive in dim light; operates at high luminance), high visual acuity, most sensitive to long wavelengths, basis of colour vision

- **Scotopic vision**

Rod-dominated, peripheral vision, dark-adapted vision (sensitive in dim light; operates at low luminance), low visual acuity, sensitive to short wavelengths, no colour sensation

(**Mesopic vision** = rod and cone vision together)

Summary

- We have looked at
 - Eye **anatomy**.
 - Visible light as the **basis of vision**.
 - The **1st transformation** of the visual process:
 - From light to retinal object images
 - How objects are put into focus (**accommodation**)
 - The **2nd transformation** of the visual process
 - Transduction from a retinal images to electrical signals
 - How different **photo receptors** shape vision in different ways

Test yourself – learning goals

- After reviewing this lecture you should be able to
 - Name the two most important landmarks on the retina and describe why they are important.
 - Describe what the result of the 1st transformation is.
 - Explain the process of accommodation and what the far and near point are.
 - Explain Myopia and Hyperopia.
 - Describe the structure of the retina.
 - Describe what the result of the 2nd transformation is.
 - Describe the 7 most important differences between rods and cones and how they lead to two different types of vision.

Test yourself – multiple choice

1. Which of the following statements about neural convergence is correct?
 - a) Neural convergence is beneficial for scotopic vision, as it increases the sensitivity of rods.
 - b) Neural convergence is beneficial for photopic vision, as it increases visual acuity.
 - c) Neural convergence is beneficial for both scotopic and photopic vision, as it affects rods and cones to the same extend.
 - d) Neural convergence is neither beneficial for photopic nor for scotopic vision, as it only a descriptive construct.

Solutions will be discussed on Friday.

Test yourself – multiple choice

2. Which of the following statements leads to an erroneous sentence? One of the landmarks on the retina is the optic disc, ...

- a) ...this is where the optic nerve leaves the eye.
- b) ...this is where the axons of the ganglion cells leave the eye.
- c) ...this is where visual acuity is highest.
- d) ...this is where the retina has no receptors.

Test yourself – multiple choice

3. How does accommodation to a near object work?

- a) The ciliary muscles relax to make the lens less curvy. This will bend the light waves only a little and they fall directly onto the fovea.
- b) The ciliary muscles relax to make the lens curvy. This will cause the light waves to be bent a lot and they fall directly onto the fovea.
- c) The ciliary muscles tighten to make the lens curvy. This will cause the light waves to be bent a little and they fall directly onto the fovea.
- d) The ciliary muscles tighten to make the lens curvy. This will cause the light waves to be bent a lot and they fall directly onto the fovea.

See you on Friday!

